

Note that we will be using Green's theorem only for relatively simple domains, those whose boundaries consist of straight line segments and circular arcs, for which Green's theorem can be established relatively easily.

Exercises for III.1

1. Evaluate $\int_{\gamma} y^2 dx + x^2 dy$ along the following paths γ from $(0, 0)$ to $(2, 4)$: (a) the arc of the parabola $y = x^2$; (b) the horizontal interval from $(0, 0)$ to $(2, 0)$, followed by the vertical interval from $(2, 0)$ to $(2, 4)$; (c) the vertical interval from $(0, 0)$ to $(0, 4)$, followed by the horizontal interval from $(0, 4)$ to $(2, 4)$.
2. Evaluate $\int_{\gamma} xy dx$ both directly and using Green's theorem, where γ is the boundary of the square with vertices at $(0, 0)$, $(1, 0)$, $(1, 1)$, and $(0, 1)$.
3. Evaluate $\int_{\partial D} x^2 dy$ both directly and using Green's theorem, where D is the quarter-disk in the first quadrant bounded by the unit circle and the two coordinate axes.
4. Evaluate $\int_{\gamma} y dx$ both directly and using Green's theorem, where γ is the semicircle in the upper half-plane from R to $-R$.
5. Show that $\int_{\partial D} x dy$ is the area of D , while $\int_{\partial D} y dx$ is minus the area of D .
6. Show that if P and Q are continuous complex-valued functions on a curve γ , then

$$F(w) = \int_{\gamma} \frac{P dx}{z - w} + \int_{\gamma} \frac{Q dy}{z - w} \quad (z = x + iy)$$

is analytic for $w \in \mathbb{C} \setminus \gamma$. Express $F'(w)$ as a line integral over γ .

7. Show that the formula in Green's theorem is invariant under coordinate changes, in the sense that if the theorem holds for a bounded domain U with piecewise smooth boundary, and if $F(x, y)$ is a smooth function that maps U one-to-one onto another such domain V and that maps the boundary of U one-to-one smoothly onto the boundary of V , then Green's theorem holds for V . *Hint.* First note the change of variable formulae for line and area integrals, given by

$$\begin{aligned} \int_{\partial V} P d\xi &= \int_{\partial U} (P \circ F) \left(\frac{\partial \xi}{\partial x} dx + \frac{\partial \xi}{\partial y} dy \right), \\ \iint_V R d\xi d\eta &= \iint_U (R \circ F) \det J_F dx dy, \end{aligned}$$

where $F(x, y) = (\xi(x, y), \eta(x, y))$, and where J_F is the Jacobian matrix of F . Use these formulae, with $R = -\partial P/\partial \eta$. The summand $\int Q d\eta$ is treated similarly.

8. Prove Green's theorem for the rectangle defined by $x_0 < x < x_1$ and $y_0 < y < y_1$ (a) directly, and (b) using the result for triangles.

2. Independence of Path

In order to draw a useful analogy with single-variable calculus, we begin by reviewing the fundamental theorem of calculus. Recall that $F(t)$ is an **antiderivative** for $f(t)$ if its derivative is f , that is, $F' = f$.

Fundamental Theorem of Calculus.

Part I. If $F(t)$ is an antiderivative for the continuous function $f(t)$, then

$$\int_a^b f(t) dt = F(b) - F(a).$$

Part II. If $f(t)$ is a continuous function on $[a, b]$, then the indefinite integral

$$F(t) = \int_a^t f(s) ds, \quad a \leq t \leq b,$$

is an antiderivative for $f(t)$. Further, each antiderivative for $f(t)$ differs from $F(t)$ by a constant.

If $h(x, y)$ is a continuously differentiable complex-valued function, we define the **differential** dh of h by

$$dh = \frac{\partial h}{\partial x} dx + \frac{\partial h}{\partial y} dy.$$

We say that a differential $Pdx + Qdy$ is **exact** if $Pdx + Qdy = dh$ for some function h . The function h plays the role of the antiderivative, and the following theorem is the analogue of Part I of the fundamental theorem of calculus. It provides a useful tool for evaluating line integrals.

Theorem (Part I). If γ is a piecewise smooth curve from A to B , and if $h(x, y)$ is continuously differentiable on γ , then

$$(2.1) \quad \int_{\gamma} dh = h(B) - h(A).$$

To see this, let the curve be given by $t \mapsto (x(t), y(t))$, $a \leq t \leq b$. From (1.3) we have

$$\int_{\gamma} dh = \int_{\gamma} \frac{\partial h}{\partial x} dx + \frac{\partial h}{\partial y} dy = \int_a^b \frac{\partial h}{\partial x} \frac{dx}{dt} dt + \int_a^b \frac{\partial h}{\partial y} \frac{dy}{dt} dt.$$

By the chain rule and the fundamental theorem of calculus, this is

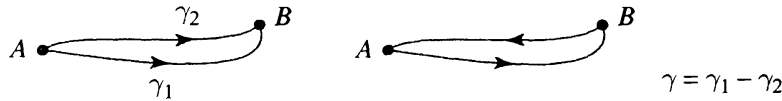
$$\int_a^b \frac{d}{dt} h(x(t), y(t)) dt = h(x(t), y(t)) \Big|_a^b = h(B) - h(A).$$

Example. To evaluate $\int_{\gamma} 2xy dx + (x^2 + 2y) dy$, where γ is the quarter-circle given by $\gamma(\theta) = (\cos \theta, \sin \theta)$, $0 \leq \theta \leq \pi/2$, we could proceed as in the preceding section and plug the parameterizing functions into (1.3). However, in this case it is easier to observe that $2xy dx + (x^2 + 2y) dy = dh$ for $h(x, y) = x^2 y + y^2$. Consequently,

$$\int_{\gamma} 2xy dx + (x^2 + 2y) dy = (x^2 y + y^2) \Big|_{(1,0)}^{(0,1)} = 1 - 0 = 1.$$

Unfortunately, not every differential $Pdx + Qdy$ is exact. We aim to give some conditions that can be used to determine when a differential is exact.

Let P and Q be continuous complex-valued functions on a domain D . We say that the line integral $\int Pdx + Qdy$ is **independent of path** in D if for any two points A and B of D , the integrals $\int_{\gamma} Pdx + Qdy$ are the same for any path γ in D from A to B . This is tantamount to requiring $\int_{\gamma} Pdx + Qdy = 0$ for any closed path γ in D . Indeed, if γ_1 and γ_2 are two paths in D from A to B , then we can form a closed path γ in D , starting and ending at A , by following γ_1 from A to B and then following γ_2 backwards from B to A . Since the reversal of direction along γ_2 changes the sign of the integral, we have $\int_{\gamma} = \int_{\gamma_1} - \int_{\gamma_2}$, so that $\int_{\gamma} = 0$ if and only if $\int_{\gamma_1} = \int_{\gamma_2}$.



Formula (2.1) shows that the integrals of exact differentials are independent of path. The converse is easily seen to be true also.

Lemma. Let P and Q be continuous complex-valued functions on a domain D . Then $\int Pdx + Qdy$ is independent of path in D if and only if $Pdx + Qdy$ is exact, that is, there is a continuously differentiable function $h(x, y)$ such that $dh = Pdx + Qdy$. Moreover, the function h is unique, up to adding a constant.

Suppose that $\int Pdx + Qdy$ is independent of path in D . Fix a point A in D , and define a function $h(x, y)$ on D by

$$h(B) = \int_A^B P dx + Q dy, \quad B \in D,$$

where we may take any path in D from A to B . We compute the partial derivatives of $h(x, y)$ by choosing some special paths. Fix (x_0, y_0) in D ,

and fix a path γ from A to (x_0, y_0) . For x near x_0 , we evaluate $h(x, y_0)$ by following the path γ from A to (x_0, y_0) and then the straight line path $x(t) = t, y(t) = y_0$ from (x_0, y_0) to (x, y_0) . This gives

$$h(x, y_0) = \int_{\gamma} Pdx + Qdy + \int_{x_0}^x P(t, y_0) dt.$$

Since the first summand on the right is a constant, we obtain from the fundamental theorem of calculus that

$$\frac{\partial h}{\partial x}(x_0, y_0) = P(x_0, y_0).$$

Similarly, we obtain

$$\frac{\partial h}{\partial y}(x_0, y_0) = Q(x_0, y_0),$$

and consequently $dh = Pdx + Qdy$. For the uniqueness, note that if h_1 is any other function such that $dh_1 = Pdx + Qdy$, then $d(h - h_1) = 0$, that is,

$$\frac{\partial}{\partial x}(h - h_1) = 0 = \frac{\partial}{\partial y}(h - h_1).$$

Since D is a domain, $h - h_1$ is constant on D .

Let P and Q be continuously differentiable complex-valued functions on a domain D . We say that the differential $Pdx + Qdy$ is **closed** on D if

$$(2.2) \quad \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

This is precisely the condition that the integrand in Green's theorem is zero. Thus Green's theorem implies that if $Pdx + Qdy$ is closed on D , then $\int_{\partial U} Pdx + Qdy = 0$ for any bounded domain U with piecewise smooth boundary such that U together with its boundary is contained in D .

Lemma. *Exact differentials are closed.*

Indeed, if $Pdx + Qdy = dh$ is exact, then

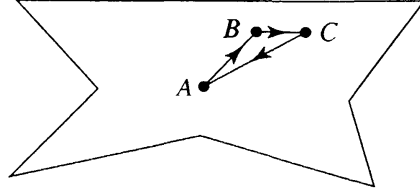
$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y} \frac{\partial h}{\partial x} = \frac{\partial}{\partial x} \frac{\partial h}{\partial y} = \frac{\partial Q}{\partial x}.$$

Not every closed differential is exact. For certain domains, the so-called simply connected domains, any closed differential is exact; in fact, this statement characterizes simply connected domains. We content ourselves with the following theorem for star-shaped domains, which includes the simply connected domains of most interest to us. The theorem is the analogue of Part II of the fundamental theorem of calculus. It gives conditions on a smooth differential to have an antiderivative, that is, to be exact.

Theorem (Part II). *Let P and Q be continuously differentiable complex-valued functions on a domain D . Suppose*

- (i) *D is a star-shaped domain (as a disk or rectangle), and*
- (ii) *the differential $Pdx + Qdy$ is closed on D .*

Then $Pdx + Qdy$ is exact on D .



The proof is similar to the one given just above. Suppose that D is star-shaped with respect to the point $A \in D$. We define $h(B)$ at any point $B \in D$ by

$$h(B) = \int_A^B P dx + Q dy,$$

where the path of integration is the straight line segment from A to B . We claim that $dh = Pdx + Qdy$. To see this, fix $B = (x_0, y_0)$, and let $C = (x, y_0)$ lie on the horizontal line through B and close enough to B so that the triangle with vertices A, B, C lies within D . We apply Green's theorem to the triangle, to obtain

$$\left(\int_A^B + \int_B^C + \int_C^A \right) (P dx + Q dy) = 0.$$

Thus

$$\int_A^C (P dx + Q dy) - \int_A^B (P dx + Q dy) = \int_B^C (P dx + Q dy),$$

or

$$h(x, y_0) - h(x_0, y_0) = \int_{x_0}^x P(t, y_0) dt.$$

From the fundamental theorem of calculus we obtain

$$\frac{\partial h}{\partial x}(x_0, y_0) = P(x_0, y_0).$$

Similarly,

$$\frac{\partial h}{\partial y}(x_0, y_0) = Q(x_0, y_0).$$

Consequently, $dh = Pdx + Qdy$, and $Pdx + Qdy$ is exact.

Example. Consider the differential

$$\frac{-y dx + x dy}{x^2 + y^2}, \quad x + iy \in \mathbb{C} \setminus \{0\}.$$

A straightforward calculation (Exercise 2) reveals that the differential is closed on $\mathbb{C} \setminus \{0\}$. If we integrate the differential around the unit circle, using the parameterization $x = \cos \theta$, $y = \sin \theta$, we obtain

$$\oint_{|z|=1} \frac{-y dx + x dy}{x^2 + y^2} = \int_0^{2\pi} d\theta = 2\pi.$$

Thus the integral is not independent of path, and the differential is not exact on $\mathbb{C} \setminus \{0\}$. On the other hand, since $\mathbb{C} \setminus (-\infty, 0]$ is a star-shaped domain, the differential *is* exact on $\mathbb{C} \setminus (-\infty, 0]$. On this domain the differential coincides with $d(\text{Arg } z)$.

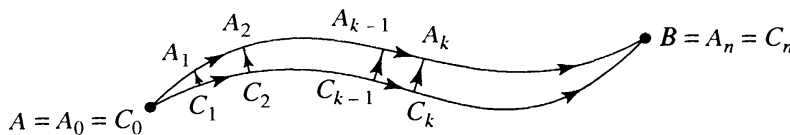
Now suppose that $Pdx + Qdy$ is a closed differential on a domain D . We fix points $A, B \in D$, and we consider paths γ in D from A to B . The integral $\int_{\gamma} Pdx + Qdy$ may depend on the path γ . We claim, however, that if two paths γ_0 and γ are “sufficiently near” to each other, then $\int_{\gamma_0} Pdx + Qdy = \int_{\gamma} Pdx + Qdy$. By “sufficiently near” we mean that there are successive points $A = A_0, A_1, A_2, \dots, A_n = B$ on γ_0 and $A = C_0, C_1, C_2, \dots, C_n = B$ on γ such that the intervals on γ_0 from A_{k-1} to A_k and on γ from C_{k-1} to C_k are contained in the same disk Δ_k , which is contained in D . To see that the integrals are the same, we let γ_k be the path in D that follows γ from A to the point C_k , then follows a straight line segment in Δ_k from C_k to A_k , then follows γ_0 from A_k to B . Thus γ_k is obtained from γ_{k-1} by changing only the subpath in Δ_k from C_{k-1} to A_k , so that instead of following the straight line from C_{k-1} to A_{k-1} and then γ_0 from A_{k-1} to A_k , we follow γ from C_{k-1} to C_k and then the straight line from C_k to A_k . Since the integral of $Pdx + Qdy$ is independent of path in the disk Δ_k , this change in γ_{k-1} to γ_k does not affect the integral,

$$\int_{\gamma_{k-1}} Pdx + Qdy = \int_{\gamma_k} Pdx + Qdy, \quad 1 \leq k \leq n.$$

Since $\gamma_n = \gamma$, we obtain after n steps

$$\int_{\gamma} Pdx + Qdy = \int_{\gamma_0} Pdx + Qdy.$$

This identity holds not only if γ is near γ_0 but whenever γ can be obtained by deforming γ_0 continuously. We state the deformation theorem formally.

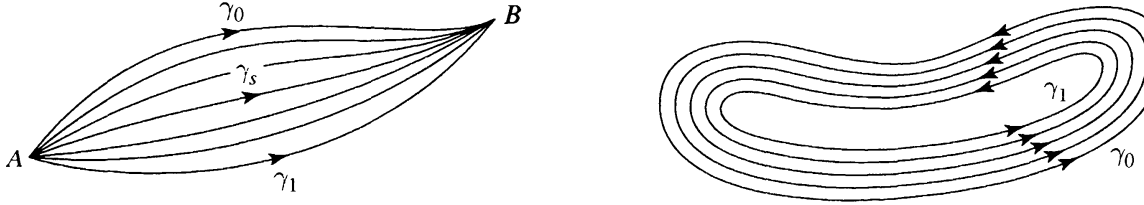


Theorem. Let D be a domain, and let $\gamma_0(t)$ and $\gamma_1(t)$, $a \leq t \leq b$, be two paths in D from A to B . Suppose that γ_0 can be continuously deformed to γ_1 , in the sense that for $0 \leq s \leq 1$ there are paths $\gamma_s(t)$, $a \leq t \leq b$, from A to B such that $\gamma_s(t)$ depends continuously on s and t for $0 \leq s \leq 1$, $a \leq t \leq b$. Then

$$(2.3) \quad \int_{\gamma_0} Pdx + Qdy = \int_{\gamma_1} Pdx + Qdy$$

for any closed differential $Pdx + Qdy$ on D .

The idea of the proof is that any continuous deformation of γ_0 can be realized as a finite number of successive modifications by paths in D from A to B that are close to each other. The proof requires a compactness argument, but otherwise it is straightforward. The compactness argument allows us to find paths corresponding to $0 = s_0 < s_1 < \cdots < s_m = 1$, and successive t -values $a = t_0 < t_1 < \cdots < t_n = b$ such that the subpaths $\gamma_{s_{j-1}}(t)$ and $\gamma_{s_j}(t)$, $t_{k-1} \leq t \leq t_k$, lie in the same disk Δ_{jk} in D . This can be done on account of the continuity of $\gamma_s(t)$ in s and t . Then $\gamma_{s_{j-1}}$ is “sufficiently near” to γ_{s_j} for each j , and the rolling wave argument above shows that the integrals of $Pdx + Qdy$ over the two paths are the same, this for $1 \leq j \leq n$, so that (2.3) holds.



A slight variation of the argument establishes a deformation theorem for closed paths in D . These are paths in D that start and end at the same point. When we deform closed paths, we allow the starting point to move also.

Theorem. Let D be a domain, and let $\gamma_0(t)$ and $\gamma_1(t)$, $a \leq t \leq b$, be two closed paths in D . Suppose that γ_0 can be continuously deformed to γ_1 , in the sense that for $0 \leq s \leq 1$ there are closed paths $\gamma_s(t)$, $a \leq t \leq b$, such that $\gamma_s(t)$ depends continuously on s and t for $0 \leq s \leq 1$, $a \leq t \leq b$. Then

$$\int_{\gamma_0} Pdx + Qdy = \int_{\gamma_1} Pdx + Qdy$$

for any closed differential $Pdx + Qdy$ on D .

Summary. We have defined what it means for a differential $Pdx + Qdy$ to be **exact**, to be **closed**, and to be **independent of path**. We have

shown that

$$\text{independent of path} \iff \text{exact} \implies \text{closed}.$$

For star-shaped domains we have shown that

$$\text{independent of path} \iff \text{exact} \iff \text{closed}.$$

We have also shown that if $Pdx + Qdy$ is a closed differential, then a deformation in the path from A to B does not change the value of the integral of $Pdx + Qdy$ along the path.

Exercises for III.2

1. Determine whether each of the following line integrals is independent of path. If it is, find a function h such that $dh = Pdx + Qdy$. If it is not, find a closed path γ around which the integral is not zero. (a) $x dx + y dy$, (b) $x^2 dx + y^5 dy$, (c) $y dx + x dy$, (d) $y dx - x dy$.
2. Show that the differential

$$\frac{-y dx + x dy}{x^2 + y^2}, \quad (x, y) \neq (0, 0),$$

is closed. Show that it is not independent of path on any annulus centered at 0.

3. Suppose that P and Q are smooth functions on the annulus $\{a < |z| < b\}$ that satisfy $\partial P / \partial y = \partial Q / \partial x$. Show directly using Green's theorem that $\oint_{|z|=r} P dx + Q dy$ is independent of the radius r , for $a < r < b$.
4. Let P and Q be smooth functions on D satisfying $\partial P / \partial y = \partial Q / \partial x$. Let γ_0 and γ_1 be two closed paths in D such that the straight line segment from $\gamma_0(t)$ to $\gamma_1(t)$ lies in D for every parameter value t . Then $\int_{\gamma_0} P dx + Q dy = \int_{\gamma_1} P dx + Q dy$. Use this to give another solution to the preceding exercise.
5. Let $\gamma_0(t)$ and $\gamma_1(t)$, $0 \leq t \leq 1$, be paths in the slit annulus $\{a < |z| < b\} \setminus (-b, -a)$ from A to B . Write down explicitly a family of paths $\gamma_s(t)$ from A to B in the slit annulus that deforms γ_0 continuously to γ_1 . *Suggestion.* Deform separately the modulus and the principal value of the argument.
6. Show that any closed path $\gamma(t)$, $0 \leq t \leq 1$, in the annulus $\{a < |z| < b\}$ can be deformed continuously to the circular path $\sigma(t) = \gamma(0)e^{2\pi i m t}$, $0 \leq t \leq 1$, for some integer m . *Hint.* Reduce to the case where $|\gamma(t)| \equiv |\gamma(0)|$ is constant. Then start by finding a subdivision $0 = t_0 < t_1 < \cdots < t_n = 1$ such that $\arg \gamma(t)$ has a continuous determination on each interval $t_{j-1} \leq t \leq t_j$.

7. Show that if 0 and ∞ lie in different connected components of the complement $\mathbb{C}^* \setminus D$ of D in the extended complex plane, then there is a closed path γ in D such that $\int_{\gamma} d\theta \neq 0$. *Hint.* The hypothesis means that there are $\delta > 0$ and a bounded subset E of $\mathbb{C} \setminus D$ such that $0 \in E$, and every point of E has distance at least 5δ from every point of $\mathbb{C} \setminus D$ not in E . Lay down a grid of squares in the plane with side length δ , and let F be the union of the closed squares in the grid that meet E or that border on a square meeting E . Show that ∂F is a finite union of closed paths in D , and that $\int_{\partial F} d\theta = 2\pi$.

3. Harmonic Conjugates

The basis for application of Green's theorem to harmonic functions is the following important observation.

Lemma. *If $u(x, y)$ is harmonic, then the differential*

$$(3.1) \quad -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$$

is closed.

Indeed, for this differential the condition (2.1) for $P = -\partial u/\partial y$ and $Q = \partial u/\partial x$ becomes $-\partial^2 u/\partial y^2 = \partial^2 u/\partial x^2$, which is equivalent to Laplace's equation.

Now suppose that $u(x, y)$ is harmonic on a star-shaped domain D . If we apply the theorem in Section 2 to the differential given in (3.1), we obtain a smooth function $v(x, y)$ such that

$$(3.2) \quad dv = -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy.$$

The equation (3.2) is equivalent to

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \quad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x},$$

which are the Cauchy-Riemann equations. Thus $u + iv$ is analytic, and we have established the following theorem.

Theorem. *Any harmonic function $u(x, y)$ on a star-shaped domain D (as a disk or rectangle) has a harmonic conjugate function $v(x, y)$ on D .*

By (3.2), the harmonic conjugate $v(x, y)$ is given explicitly up to an additive constant by

$$(3.3) \quad v(B) = \int_A^B -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy,$$

where A is fixed, and the integral is independent of path in D . If D is a disk, and we take the path from A to B to be a vertical interval followed by a horizontal interval, we obtain the formula (5.3) derived in Section II.5.

Example. To find a harmonic conjugate $v(z)$ for $u = \log |z|$ on the star-shaped domain $\mathbb{C} \setminus (-\infty, 0]$, we express u in the form

$$u(x, y) = \frac{1}{2} \log(x^2 + y^2),$$

and we compute

$$du = \frac{x}{x^2 + y^2} dx + \frac{y}{x^2 + y^2} dy.$$

Equation (3.2) becomes

$$dv = -\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

This leads to the identity

$$\operatorname{Arg} z = \int_1^z \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy, \quad z \notin (-\infty, 0],$$

since the principal branch $\operatorname{Arg} z$ is the unique harmonic conjugate of $\log |z|$ on $\mathbb{C} \setminus (-\infty, 0]$, normalized to vanish at $z = 1$.

Exercises for III.3

- For each of the following harmonic functions u , find du , find dv , and find v , the conjugate harmonic function of u .

(a) $u(x, y) = x - y$	(c) $u(x, y) = \sinh x \cos y$
(b) $u(x, y) = x^3 - 3xy^2$	(d) $u(x, y) = \frac{y}{x^2 + y^2}$
- Show that a complex-valued function $h(z)$ on a star-shaped domain D is harmonic if and only if $h(z) = f(z) + \overline{g(z)}$, where $f(z)$ and $g(z)$ are analytic on D .
- Let $D = \{a < |z| < b\} \setminus (-b, -a)$, an annulus slit along the negative real axis. Show that any harmonic function on D has a harmonic conjugate on D . *Suggestion.* Fix c between a and b , and define $v(z)$ explicitly as a line integral along the path consisting of the straight line from c to $|z|$ followed by the circular arc from $|z|$ to z . Or map the slit annulus to a rectangle by $w = \operatorname{Log} z$.
- Let $u(z)$ be harmonic on the annulus $\{a < |z| < b\}$. Show that there is a constant C such that $u(z) - C \log |z|$ has a harmonic conjugate on the annulus. Show that C is given by

$$C = \frac{r}{2\pi} \int_0^{2\pi} \frac{\partial u}{\partial r} (re^{i\theta}) d\theta,$$

where r is any fixed radius, $a < r < b$.

5. The flux of a function u across a curve γ is defined to be

$$\int_{\gamma} \frac{\partial u}{\partial n} ds = \int_{\gamma} \nabla u \cdot \mathbf{n} ds,$$

where $\mathbf{n}$ is the unit normal vector to γ and ds is arc length. Show that if a harmonic function u on a domain D has a conjugate harmonic function v on D , then the integral giving the flux is independent of path in D . Further, the flux across a path γ in D from A to B is $v(B) - v(A)$.

4. The Mean Value Property

Let $h(z)$ be a continuous real-valued function on a domain D . Let $z_0 \in D$, and suppose D contains the disk $\{|z - z_0| < \rho\}$. We define the **average value** of $h(z)$ on the circle $\{|z - z_0| = r\}$ to be

$$A(r) = \int_0^{2\pi} h(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}, \quad 0 < r < \rho.$$

Since $h(z)$ is continuous, the average value $A(r)$ varies continuously with the radius r . Since the values of $h(z)$ are all near $h(z_0)$ when z is near z_0 , the averages $A(r)$ are also near $h(z_0)$ when r is small, and consequently $A(r)$ tends to $h(z_0)$ as r decreases to 0. This can be seen with complete rigor from the estimates

$$\begin{aligned} |A(r) - h(z_0)| &= \left| \int_0^{2\pi} [h(z_0 + re^{i\theta}) - h(z_0)] \frac{d\theta}{2\pi} \right| \\ &\leq \int_0^{2\pi} |h(z_0 + re^{i\theta}) - h(z_0)| \frac{d\theta}{2\pi}, \end{aligned}$$

where we have used the fact that $d\theta/2\pi$ is a probability measure, that is, it is positive and its integral is 1. The continuity of $h(z)$ at z_0 guarantees that the integrand in the right-hand side tends to 0 uniformly in θ , so that the integral tends to 0 as r tends to 0.

Theorem. *If $u(z)$ is a harmonic function on a domain D , and if the disk $\{|z - z_0| < \rho\}$ is contained in D , then*

$$(4.1) \quad u(z_0) = \int_0^{2\pi} u(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}, \quad 0 < r < \rho.$$

In other words, the average value of a harmonic function on the boundary circle of any disk contained in D is its value at the center of the disk. To

see this, we begin with the identity

$$0 = \oint_{|z-z_0|=r} -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy,$$

which follows immediately from Green's theorem and the harmonicity of $u(z)$. (See the Lemma in Section 3.) We parametrize the circle by $x(\theta) = x_0 + r \cos \theta$ and $y(\theta) = y_0 + r \sin \theta$, and we obtain

$$(4.2) \quad 0 = r \int_0^{2\pi} \left[\frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \right] d\theta = r \int_0^{2\pi} \frac{\partial u}{\partial r} (z_0 + re^{i\theta}) d\theta.$$

Since $u(z)$ is smooth, we can interchange the order of integration and differentiation. We obtain after dividing by $2\pi r$ that

$$0 = \frac{\partial}{\partial r} \int_0^{2\pi} u(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}, \quad 0 < r < \rho.$$

Thus

$$\int_0^{2\pi} u(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}$$

is constant for $0 < r < \rho$. Since $u(z)$ is continuous at z_0 , the average value tends to $u(z_0)$ as $r \rightarrow 0$, and the constant is $u(z_0)$. This establishes (4.1).

We say that a continuous function $h(z)$ on a domain D has the **mean value property** if for each point $z_0 \in D$, $h(z_0)$ is the average of its values over any small circle centered at z_0 . In the language of formal mathematics, this means that for any $z_0 \in D$, there is $\varepsilon > 0$ such that

$$h(z_0) = \int_0^{2\pi} h(z_0 + re^{i\theta}) \frac{d\theta}{2\pi}, \quad 0 < r < \varepsilon.$$

Our theorem above has a simple restatement: Harmonic functions have the mean value property. We will show in Chapter X that the converse is true, that any continuous function on D with the mean value property is harmonic. This is rather remarkable, since the hypothesis requires only continuity but no differentiability of the function.

Exercises for III.4

1. Let $f(z)$ be a continuous function on a domain D . Show that if $f(z)$ has the mean value property with respect to circles, as defined above, then $f(z)$ has the mean value property with respect to disks, that is, if $z_0 \in D$ and D_0 is a disk centered at z_0 with area A and contained in D , then $f(z_0) = \frac{1}{A} \iint_{D_0} f(z) dx dy$.
2. Derive (4.2) from the polar form of the Cauchy-Riemann equations (Exercise II.3.8).

3. A function $f(t)$ on an interval $I = (a, b)$ has the **mean value property** if

$$f\left(\frac{s+t}{2}\right) = \frac{f(s) + f(t)}{2}, \quad s, t \in I.$$

Show that any affine function $f(t) = At + B$ has the mean value property. Show that any continuous function on I with the mean value property is affine.

4. Formulate the mean value property for a function on a domain in $\mathbb{R}^3$, and show that any harmonic function has the mean value property. *Hint.* For $A \in \mathbb{R}^3$ and $r > 0$, let B_r be the ball of radius r centered at A , with volume element $d\tau$, and let ∂B_r be its boundary sphere, with area element $d\sigma$ and unit outward normal vector $\mathbf{n}$. Apply the Gauss divergence theorem

$$\iint_{\partial B_r} \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_{B_r} \nabla \cdot \mathbf{F} \, d\tau$$

to $\mathbf{F} = \nabla u$.

5. The Maximum Principle

The strict maximum principle asserts that if a real-valued harmonic function attains its maximum on a domain D , then it is constant.

Strict Maximum Principle (Real Version). *Let $u(z)$ be a real-valued harmonic function on a domain D such that $u(z) \leq M$ for all $z \in D$. If $u(z_0) = M$ for some $z_0 \in D$, then $u(z) = M$ for all $z \in D$.*

The idea of the proof is to use the mean value property to show that the set of points for which $u(z) = M$ is open. Indeed, suppose $u(z_1) = M$, and express the mean value equality (4.1) in the form

$$(5.1) \quad 0 = \int_0^{2\pi} [u(z_1) - u(z_1 + re^{i\theta})] \frac{d\theta}{2\pi}, \quad 0 < r < \rho.$$

Since the integrand is nonnegative (≥ 0) and continuous, the integral (5.1) can be zero only if the integrand is zero. Thus $u(z_1 + re^{i\theta}) = u(z_1) = M$ for $0 \leq \theta \leq 2\pi$ and $0 < r < \rho$, and the set $\{u(z) = M\}$ contains a disk centered at each of its points, hence is open. Now, the set $\{u(z) < M\}$ is also open, since $u(z)$ is continuous. Since D is a domain, one of these two sets is empty and the other coincides with all of D . (See Section II.1.) In other words, either $u(z) < M$ for all $z \in D$, or $u(z) = M$ for all $z \in D$, and this proves the theorem.

Recall that a complex-valued function is **harmonic** if its real and imaginary parts are harmonic, that is, if the function satisfies Laplace's equa-

tion. Thus any analytic function is harmonic. There is a strict maximum principle also for complex-valued harmonic functions. It asserts that if a complex-valued harmonic function attains its maximum modulus on a domain D , then it is constant.

Strict Maximum Principle (Complex Version). *Let h be a bounded complex-valued harmonic function on a domain D . If $|h(z)| \leq M$ for all $z \in D$, and $|h(z_0)| = M$ for some $z_0 \in D$, then $h(z)$ is constant on D .*

This can be derived easily from the real version of the strict maximum principle. We replace $h(z)$ by $\lambda h(z)$ for an appropriate unimodular constant λ , and we can assume that $h(z_0) = M$. Let $u(z) = \operatorname{Re} h(z)$. Then $u(z)$ is a harmonic function on D that attains its maximum at z_0 . By the strict maximum principle for real-valued harmonic functions, $u(z) = M$ for all $z \in D$. Since $|h(z)| \leq M$ and $\operatorname{Re} h(z) = M$, we must have $\operatorname{Im} h(z) = 0$ for $z \in D$. Hence $h(z)$ is constant on D .

As a corollary of the strict maximum principle we also obtain the following version of the maximum principle. In words, it asserts that a complex-valued harmonic function on a bounded domain attains its maximum modulus on the boundary.

Maximum Principle. *Let $h(z)$ be a complex-valued harmonic function on a bounded domain D such that $h(z)$ extends continuously to the boundary ∂D of D . If $|h(z)| \leq M$ for all $z \in \partial D$, then $|h(z)| \leq M$ for all $z \in D$.*

The proof of the maximum principle hinges on the fact that a continuous function on a compact set attains its maximum modulus at some point of the set. (See Section II.1.) In this case the compact set is the union of the domain and its boundary, which is a closed bounded set. If the harmonic function attains its maximum modulus at some point of D , then it is constant. Thus in all cases it attains its maximum modulus on the boundary of D .

The maximum principle is useful, for instance, for demonstrating convergence of a sequence of harmonic functions. To show that a sequence of harmonic functions converges in a disk or rectangle, it suffices to obtain good estimates on the boundary of the disk or rectangle, since the boundary estimates automatically persist in the interior.

Exercises for III.5

1. Let D be a bounded domain, and let u be a real-valued harmonic function on D that extends continuously to the boundary ∂D . Show that if $a \leq u \leq b$ on ∂D , then $a \leq u \leq b$ on D .

2. Fix $n \geq 1$, $r > 0$, and $\lambda = \rho e^{i\varphi}$. What is the maximum modulus of $z^n + \lambda$ over the disk $\{|z| \leq r\}$? Where does $z^n + \lambda$ attain its maximum modulus over the disk?
3. Use the maximum principle to prove the fundamental theorem of algebra, that any polynomial $p(z)$ of degree $n \geq 1$ has a zero, by applying the maximum principle to $1/p(z)$ on a disk of large radius.
4. Let $f(z)$ be an analytic function on a domain D that has no zeros on D . (a) Show that if $|f(z)|$ attains its minimum on D , then $f(z)$ is constant. (b) Show that if D is bounded, and if $f(z)$ extends continuously to the boundary ∂D of D , then $|f(z)|$ attains its minimum on ∂D .
5. Let $f(z)$ be a bounded analytic function on the right half-plane. Suppose that $f(z)$ extends continuously to the imaginary axis and satisfies $|f(iy)| \leq M$ for all points iy on the imaginary axis. Show that $|f(z)| \leq M$ for all z in the right half-plane. *Hint.* For $\varepsilon > 0$ small, consider $(z + 1)^{-\varepsilon} f(z)$ on a large semidisk.
6. Let $f(z)$ be a bounded analytic function on the right half-plane. Suppose that $\limsup |f(z)| \leq M$ as z approaches any point of the imaginary axis. Show that $|f(z)| \leq M$ for all z in the half-plane. *Remark.* This is a technical improvement on the preceding exercise for students who can deal with a $\limsup$ (see Section V.1).
7. Let $f(z)$ be a bounded analytic function on the open unit disk $\mathbb{D}$. Suppose there are a finite number of points on the boundary such that $f(z)$ extends continuously to the arcs of $\partial\mathbb{D}$ separating the points and satisfies $|f(e^{i\theta})| \leq M$ there. Show that $|f(z)| \leq M$ on $\mathbb{D}$. *Hint.* In the case that there is only one exceptional point $z = 1$, consider the function $(1 - z)^\varepsilon f(z)$.
8. Let $f(z)$ be a bounded analytic function on a horizontal strip in the complex plane. Suppose that $f(z)$ extends continuously to the boundary lines of the strip and satisfies $|f(z)| \leq M$ there. Show that $|f(z)| \leq M$ for all z in the strip. *Hint.* Find a conformal map of the strip onto $\mathbb{D}$ and apply Exercise 7.
9. Let D be an unbounded domain, $D \neq \mathbb{C}$, and let $u(z)$ be a harmonic function on D that extends continuously to the boundary ∂D . Suppose that $u(z)$ is bounded below on D , and that $u(z) \geq 0$ on ∂D . Show that $u(z) \geq 0$ on D . *Hint.* Suppose $0 \in \partial D$, and consider functions of the form $u(z) + \rho \log |z|$ on $D \cap \{|z| > \varepsilon\}$.
10. Let D be a bounded domain, and let $z_0 \in \partial D$. Let $u(z)$ be a harmonic function on D that extends continuously to each boundary point of D except possibly z_0 . Suppose that $u(z)$ is bounded below

on D , and that $u(z) \geq 0$ for all $z \in \partial D$, $z \neq z_0$. Show that $u(z) \geq 0$ on D .

11. Let E be a bounded set of integer lattice points in the complex plane. A point $m + ni$ of E is an **interior point** of E if its four immediate neighbors $m \pm 1 + ni$, $m + ni \pm i$ belong to E . Otherwise, $m + ni$ is a **boundary point** of E . A function on E is **harmonic** if its value at any interior point of E is the average of its values at the four immediate neighbors. Show that a harmonic function on a bounded set of lattice points attains its maximum modulus on the boundary of the set.

6. Applications to Fluid Dynamics

We consider a fluid flow in a domain D in the plane. We think of the fluid as a collection of particles that move in the plane as time evolves. We associate with the particle at the point z its velocity vector $\mathbf{V}(z)$. The direction of $\mathbf{V}(z)$ is the direction the particle is moving, and the magnitude $|\mathbf{V}(z)|$ is its speed. We make the following assumptions on the flow.

1. The flow is independent of time, that is, the velocity vector field $\mathbf{V}(z)$ does not change with time.
2. There are no sources or sinks in D ; that is, no fluid is created or destroyed in D .
3. The flow is incompressible; that is, the density of the fluid is the same at each point of D , and it does not change with time.
4. The flow is “irrotational”; that is, there is no circulation of fluid around small circles centered in D .

We will return shortly to explain this fourth condition. First we define “flux” across a curve and “circulation” around a closed curve.

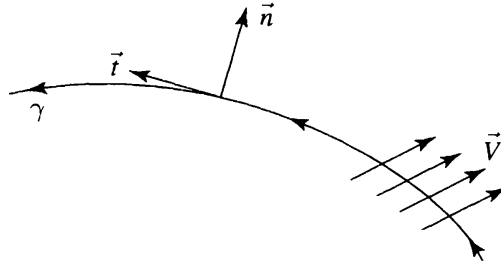
Let γ be a curve in the plane. We denote the unit tangent vector to γ by $\mathbf{t}$ and the unit normal vector to γ by $\mathbf{n}$. If γ is parameterized by arc length s , $\gamma(s) = (x(s), y(s))$, then the unit tangent and normal vectors are given respectively by

$$\mathbf{t} = \left(\frac{dx}{ds}, \frac{dy}{ds} \right), \quad \mathbf{n} = \left(\frac{dy}{ds}, -\frac{dx}{ds} \right).$$

The normal component of $\mathbf{V}$ to the curve γ is then $\mathbf{V} \cdot \mathbf{n}$. If the flow is parallel to γ , then $\mathbf{V} \cdot \mathbf{n} = 0$, and no fluid crosses γ . The maximum flow per unit length is obtained when γ is orthogonal to $\mathbf{V}$.

We define the **flux of the fluid flow** across γ to be the integral of the normal component of $\mathbf{V}$ with respect to arc length,

$$\text{flux across } \gamma = \int_{\gamma} \mathbf{V} \cdot \mathbf{n} \, ds.$$



Except for a constant involving units and the density, the flux across γ can be regarded as the amount of fluid crossing γ . If we express $\mathbf{V} = (P, Q)$ in terms of its components P and Q , we obtain an expression for the flux across γ as a line integral:

$$(6.1) \quad \int_{\gamma} \mathbf{V} \cdot \mathbf{n} \, ds = \int_{\gamma} P \frac{dy}{ds} \, ds - Q \frac{dx}{ds} \, ds = \int_{\gamma} P \, dy - Q \, dx.$$

If γ is a closed curve, we define the **circulation of the fluid flow** around γ to be the integral of the tangential component of $\mathbf{V}$ with respect to arc length:

$$\text{circulation around } \gamma = \int_{\gamma} \mathbf{V} \cdot \mathbf{t} \, ds.$$

This can also be expressed in terms of the components P and Q of $\mathbf{V}$ as a line integral,

$$(6.2) \quad \int_{\gamma} \mathbf{V} \cdot \mathbf{t} \, ds = \int_{\gamma} P \frac{dx}{ds} \, ds + Q \frac{dy}{ds} \, ds = \int_{\gamma} P \, dx + Q \, dy.$$

Let $\gamma_{\varepsilon} = z_0 + \varepsilon e^{i\theta}$ be a small circle around a fixed point $z_0 \in D$. From (6.2) and Green's theorem, the amount of fluid circulating around γ_{ε} is given by

$$\oint_{\gamma_{\varepsilon}} \mathbf{V} \cdot \mathbf{t} \, ds = \iint_{|z-z_0|<\varepsilon} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy.$$

The irrotationality of the fluid flow (the fourth condition above) means simply that this integral is zero for all small $\varepsilon > 0$. This occurs if and only if the integrand is zero, that is,

$$(6.3) \quad \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Thus the irrotationality of the flow on D means that (6.3) holds on D .

The mathematical formulation of the second and third conditions is that the net flow of fluid across the boundary of any small circle γ_{ε} centered at a point z_0 of D is zero. Using Green's theorem again, we obtain

$$0 = \oint_{\gamma_{\varepsilon}} \mathbf{V} \cdot \mathbf{n} \, ds = \int_{\gamma_{\varepsilon}} P \, dy - Q \, dx = \iint_{|z-z_0|<\varepsilon} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx \, dy.$$

We conclude as before that

$$(6.4) \quad \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0.$$

Thus the second and third conditions on the flow can be reinterpreted to assert that (6.4) holds on D .

The conditions (6.3) and (6.4) can be expressed directly in terms of the velocity $\mathbf{V}(z)$, as

$$\nabla \times \mathbf{V} = \mathbf{0}, \quad \nabla \cdot \mathbf{V} = 0.$$

This is the form of the equations that is usually most recognizable to physical scientists. It generalizes to three dimensions.

Now, the condition (6.3) is just that the differential $Pdx + Qdy$ be closed. On any disk in D , we can then find a smooth function ϕ such that $d\phi = Pdx + Qdy$, that is,

$$P = \frac{\partial \phi}{\partial x}, \quad Q = \frac{\partial \phi}{\partial y}.$$

In terms of the vector field $\mathbf{V}$, this is the same as $\nabla \phi = \mathbf{V}$. The function ϕ is called the **potential function** of $\mathbf{V}$. It is unique, up to adding a constant. In terms of ϕ , the condition (6.4) is that

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0,$$

that is, that the potential function ϕ is harmonic. On any disk in D , ϕ then has a conjugate harmonic function ψ , so that $f(z) = \phi(z) + i\psi(z)$ is analytic. The function $f(z)$ is called the **complex velocity potential** of the flow. It is also unique up to adding a constant. Note that while ϕ and ψ may be defined on any disk in D , neither ϕ nor ψ need be defined on all of D .

The velocity vector field $\mathbf{V}(z)$ is expressed in terms of the complex velocity potential $f(z) = \phi(z) + i\psi(z)$, in complex notation, by

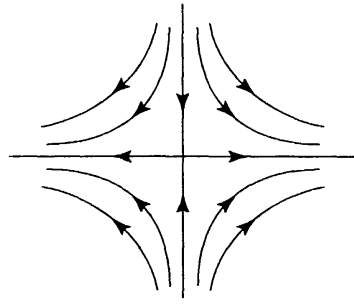
$$\mathbf{V}(z) = \frac{\partial \phi}{\partial x} + i \frac{\partial \phi}{\partial y} = \frac{\partial \phi}{\partial x} - i \frac{\partial \psi}{\partial x} = \overline{f'(z)}.$$

Thus the speed of the fluid particles is given by $|\mathbf{V}(z)| = |f'(z)|$.

As we saw in Section II.6, the level curves of ϕ and of ψ are orthogonal to each other. Since the level curves of ϕ are also orthogonal to $\nabla \phi = \mathbf{V}$, the level curves of ψ are tangent to $\mathbf{V}$. Thus the fluid particles flow along the level curves of ψ .

We define the **streamlines** of the fluid flow to be the level curves of ψ , that is, the curves $\{\psi = c\}$, for c constant. The streamlines of the flow represent the paths of the fluid particles. The function $\psi(z)$ is called the **stream function** of the fluid flow.

The stream function ψ can be used to calculate the flux of the fluid flow across a curve γ . Indeed, the flux across γ is given with the help of the

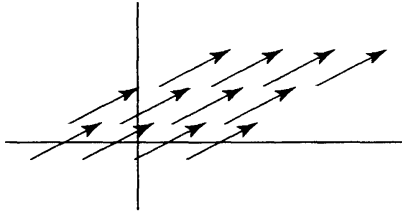


$$\begin{aligned}\vec{V}(z) &= \bar{z} = (x, -y) = \nabla\phi \\ \phi(x, y) &= (x^2 - y^2)/2 \\ \psi(z) &= xy \\ f(z) &= z^2/2\end{aligned}$$

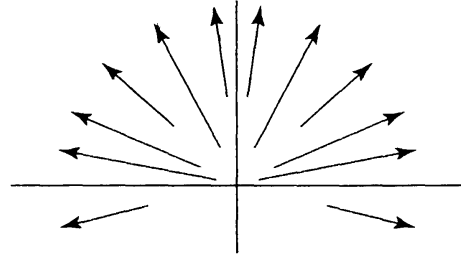
Cauchy-Riemann equations by

$$\int_{\gamma} \mathbf{V} \cdot \mathbf{n} \, ds = \int_{\gamma} \frac{\partial \phi}{\partial x} dy - \frac{\partial \phi}{\partial y} dx = \int_{\gamma} \frac{\partial \psi}{\partial y} dy + \frac{\partial \psi}{\partial x} dx = \int_{\gamma} d\psi.$$

Thus the flux of the flow across a curve γ is equal to the increase of the stream function $\psi(z)$ along γ .



$$\vec{V} = (2, 1), \phi(z) = 2x + y$$



$$\vec{V} = \frac{1}{r} \vec{u}_r, \phi(z) = \log |z|$$

Example. The simplest flow in the plane is a constant flow $\mathbf{V} = (\alpha, \beta)$, where α and β are constants. The velocity potential ϕ satisfying $\nabla\phi = \mathbf{V}$ is given, up to an additive constant, by $\phi(x, y) = \alpha x + \beta y$. The stream function ψ is the harmonic conjugate of ϕ , given up to an additive constant by $\psi(x, y) = \alpha y - \beta x$. The complex velocity potential of the flow is $f(z) = (\alpha - i\beta)z$.

Example. Consider the vector field on the punctured plane $\mathbb{C} \setminus \{0\}$ defined by

$$\mathbf{V} = \frac{1}{r} \mathbf{u}_r,$$

where $\mathbf{u}_r$ is the unit vector in the radial direction. We can express $\mathbf{V}(z)$ as $\nabla\phi(z)$ for $\phi(z) = \log |z|$. Since $\log |z|$ is harmonic, $\mathbf{V}(z)$ is the velocity vector field of a fluid flow. The stream function of the flow is $\psi(z) = \arg z$, which is defined only locally, and the complex velocity potential is $f(z) = \log z$, also defined only locally. The flux of the flow across a circle centered at the origin is calculated directly using $\mathbf{n} = \mathbf{u}_r$ and $ds = r \, d\theta$ by

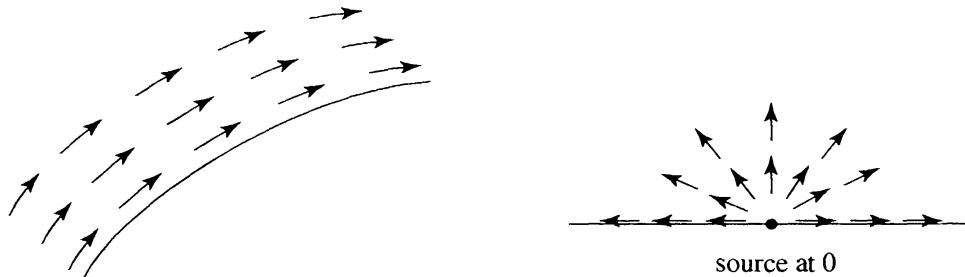
$$\oint_{|z|=r} \mathbf{V} \cdot \mathbf{n} \, ds = \int_0^{2\pi} \frac{1}{r} \mathbf{u}_r \cdot \mathbf{u}_r \, r \, d\theta = \int_0^{2\pi} d\theta = 2\pi.$$

This coincides with the increase of the stream function $\arg z$ around the circle. The origin is a source for the fluid flow. The speed of the fluid particles is given by $|\mathbf{V}(z)| = |f'(z)| = 1/|z| = 1/r$. The fluid particles emanate from the origin and follow along rays at continually diminishing speeds.

In addition to the four conditions above, there is one further important condition to be placed on a fluid flow velocity $\mathbf{V}(z)$, which is a boundary condition.

5. If no fluid is injected or extracted through a boundary curve, then the velocity vector field $\mathbf{V}(z)$ is parallel to the boundary along that curve.

Thus the boundary curve should be a streamline of the flow, and the stream function $\psi(z)$ should be constant on the curve.



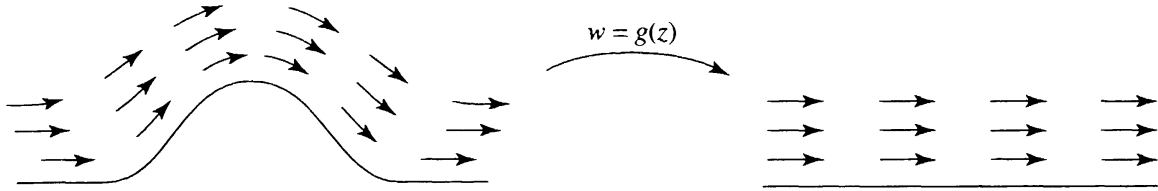
Example. We consider a fluid flow in the upper half-plane with a source or sink at 0. The fluid is injected or extracted from the upper half-plane at a constant rate at the origin. The stream function $\psi(z)$ should be a harmonic function in the upper half-plane that is constant on each of the boundary intervals $(-\infty, 0)$ and $(0, +\infty)$. Such a function is given by $\psi(z) = C \arg z$, where C is a constant. The corresponding complex velocity potential function is $f(z) = C \log z$, and the velocity vector field is $\mathbf{V}(z) = f'(z) = 1/\bar{z} = z/|z|^2$. The flux of the fluid flow entering or exiting at the origin is the increase of $\psi(z)$ along a small semicircle in the upper half-plane centered at 0, which is πC . This determines the constant C . If $C > 0$, we have a source at 0, and if $C < 0$, we have a sink at 0.

One way to gain insight into a fluid flow is to map D conformally onto a domain for which the corresponding flow is simpler to understand. Flows are preserved by conformal maps, in the following sense. If $h : D \rightarrow U$ is a one-to-one analytic function (conformal map) from D onto a domain U , and if $f_0(w) = \phi_0(w) + i\psi_0(w)$ is a complex velocity potential for a flow on U , then the composition $f(z) = f_0(h(z))$ is analytic, hence the complex velocity potential for a flow on D .

One of the simplest flows to understand is the constant horizontal flow on the upper half-plane $\mathbb{H} = \{\operatorname{Im} z > 0\}$, for which no fluid enters or leaves across the bounding real line $\mathbb{R}$. A complex velocity potential for the flow is

$f_0(z) = z$. The streamlines of the flow are horizontal lines in $\mathbb{H}$, for which $\text{Im}(z)$ is constant. The boundary $\mathbb{R}$ of $\mathbb{H}$ is a streamline, corresponding to $\text{Im}(z) = 0$.

Suppose now γ is a curve in the plane extending to infinity in both directions, and suppose D is the domain lying on one side of γ . Suppose we wish to find a fluid flow in D such that no fluid enters or exits through γ . According to the boundary condition, γ should be a streamline of the flow. The problem can be solved by finding a conformal map $h : D \rightarrow \mathbb{H}$ that maps ∞ to ∞ . In this case the complex velocity potential $f(z) = f_0(h(z))$ is simply $h(z)$ itself, and the vector velocity for the fluid flow is $\mathbf{V}(z) = \overline{h'(z)}$. The stream function is $\psi(z) = \text{Im}(h(z))$, and this is zero on γ . Thus γ is a streamline, and the boundary condition is satisfied. Note that any real multiple $Ch(z)$ is also a complex velocity potential for a fluid flow that satisfies the boundary conditions and has the same streamlines. It corresponds to multiplying the velocity vector field by C .



Example. Consider a fluid flow at a corner, represented by a sector $D = \{0 < \arg z < \alpha\}$, where $0 < \alpha \leq 2\pi$. We assume that no fluid passes through the boundary, so that the boundary of the sector should be a streamline. We map the sector conformally onto the upper half-plane by a power function

$$h(z) = z^{\pi/\alpha} = r^{\pi/\alpha} [\cos(\pi\theta/\alpha) + i \sin(\pi\theta/\alpha)].$$

We take $h(z)$ to be the complex velocity potential. Then the stream function is the imaginary part of $h(z)$, which is zero on the boundary of the sector. The streamlines are given in polar coordinates by

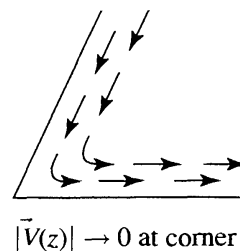
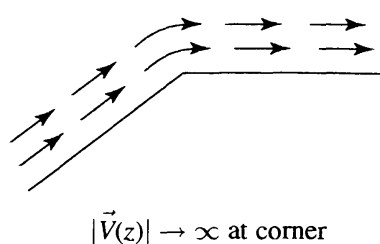
$$r = c(\sin(\pi\theta/\alpha))^{-\pi/\alpha}, \quad 0 < \theta < \alpha,$$

where c is a positive constant. The velocity vector for this flow is given by $\mathbf{V}(z) = \overline{h'(z)}$, and the speed of the flow is

$$|\mathbf{V}(z)| = |h'(z)| = \frac{\pi}{\alpha} |z|^{(\pi/\alpha)-1}.$$

If $0 < \alpha < \pi$, the fluid particles traveling near the boundary slow down as they approach the corner, while if $\pi < \alpha \leq 2\pi$, the fluid particles traveling near the boundary speed up as they make the turn around the corner.

The successful analysis of a two-dimensional flow in D using the techniques of complex analysis often depends upon being able to find a conformal map of D onto an appropriate canonical domain, such as the upper



half-plane, where the solution is apparent. In Chapter XI we will focus on enlarging our stockpile of conformal maps.

Finally, note that the flows that can be modeled as two-dimensional flows form a rather narrow class of flows. The flow of air around an airplane wing profile can be analyzed using techniques of conformal mapping, but the flow of air around an entire airplane is a three-dimensional problem to which the techniques of complex analysis do not apply.

Exercises for III.6

1. Consider the fluid flow with constant velocity $\mathbf{V} = (2, 1)$. Find the velocity potential $\phi(z)$, the stream function $\psi(z)$, and the complex velocity potential $f(z)$ of the flow. Sketch the streamlines of the flow. Determine the flux of the flow across the interval $[0, 1]$ on the real axis and across the interval $[0, i]$ on the imaginary axis.
2. Fix real numbers α and β , and consider the vector field given in polar coordinates by

$$\mathbf{V}(r, \theta) = \frac{\alpha}{r} \mathbf{u}_r + \beta \mathbf{u}_\theta,$$

where $\mathbf{u}_r$ and $\mathbf{u}_\theta$ are the unit vectors in the r and θ directions, respectively. (a) Show that $\mathbf{V}(r, \theta)$ is the velocity vector field of a fluid flow, and find the velocity potential $\phi(z)$ of the flow. (b) Find the stream function $\psi(z)$ and the complex velocity potential $f(z)$ of the flow. (c) Determine the flux of the flow emanating from the origin. When is 0 a source and when is 0 a sink? (d) Sketch the streamlines of the flow in the case $\alpha = -1$ and $\beta = 1$.

3. Consider the fluid flow with velocity $\mathbf{V} = \nabla\phi$, where $\phi(r, \theta) = (\cos\theta)/r$. Show that the streamlines of the flow are circles and sketch them. Determine the flux of the flow emanating from the origin.
4. Consider the fluid flow with velocity $\mathbf{V} = \nabla\phi$, where

$$\phi(z) = \log \left| \frac{z-1}{z+1} \right|.$$

Show that the streamlines of the flow are circles and sketch them. Determine the flux of the flow emanating from each of the singularities at ± 1 .

5. Consider the fluid flow in the horizontal strip $\{0 < \text{Im } z < \pi\}$ with a sink at 0 and equal sources at $\pm\infty$. Find the stream function $\psi(z)$ and the velocity vector field $\mathbf{V}(z)$ of the flow. Sketch the streamlines of the flow. *Hint.* Map the strip to a half-plane by $\zeta = e^z$ and solve a Dirichlet problem with constant boundary values on the three intervals in the boundary separating sinks and sources.
6. For a fluid flow with velocity potential $\phi(z)$, we define the **conjugate flow** to be the flow whose velocity potential is the conjugate harmonic function $\psi(z)$ of $\phi(z)$. What is the stream function of the conjugate flow? What is the complex velocity potential of the conjugate flow?
7. Find the stream function and the complex velocity potential of the conjugate flow associated with the fluid flow with velocity vector $\mathbf{u}_r/r$. Sketch the streamlines of the conjugate flow. Do the particles near the origin travel faster or slower than particles on the unit circle?
8. Find the stream function of the conjugate flow of

$$\mathbf{V}(r, \theta) = \frac{-\mathbf{u}_r}{r} + \mathbf{u}_\theta.$$

Sketch the streamlines of both the flow and the conjugate flow on the same axes. (See Exercise 2d.)

7. Other Applications to Physics

There are two other physical phenomena that are completely analogous to fluid dynamics. They are steady-state heat flow and electrostatics. To draw the analogy, we address each topic area briefly, beginning with steady-state heat flow.

The normalized heat equation is $u_t = \Delta u$, where u_t denotes the derivative of u with respect to time. The steady-state (or time-independent) equation for the heat is obtained by setting $u_t = 0$. The heat equation reduces to Laplace's equation $\Delta u = 0$, and thus the steady-state heat distribution is a harmonic function. In connection with steady-state heat distribution, it is natural to consider boundary-value problems of the following types.

Dirichlet Problem. Given a prescribed function v on the boundary of D , interpreted as the distribution of heat on the boundary, find a harmonic function u on D such that $u = v$ on the boundary of D .

Neumann Problem. Given a prescribed function v on the boundary of D , interpreted as the rate of flow of heat through the boundary, find a harmonic function u on D whose normal derivative $\partial u / \partial n = \nabla u \cdot \mathbf{n}$ coincides with v on the boundary of D .

Under reasonable conditions on D and v , the Dirichlet problem has a unique solution u . For a steady-state solution of the Neumann problem, there can be no net flow of heat through the boundary, $\int_{\partial D} v \, ds = 0$. Subject to this condition, the Neumann problem also has a unique solution.

The vector field $\mathbf{Q} = \nabla u$ is called the **field of flow of thermal energy**. It satisfies

$$\begin{aligned}\nabla \times \mathbf{Q} &= \nabla \times \nabla u = \mathbf{0}, \\ \nabla \cdot \mathbf{Q} &= \Delta u = 0.\end{aligned}$$

The equation $\nabla \times \mathbf{Q} = \mathbf{0}$ means that the field $\mathbf{Q}$ is **irrotational**. The equation $\nabla \cdot \mathbf{Q} = 0$ means that there are no heat sources or sinks within D . The heat flux across an arbitrary curve γ is given by

$$\int_{\gamma} \mathbf{Q} \cdot \mathbf{n} \, ds,$$

where $\mathbf{n}$ is the unit normal to γ and ds is arc length.

Exercise. Find the steady-state heat distribution in a laminar plate represented by the unit disk $\{x^2 + y^2 < 1\}$ when the boundary is held at a constant temperature $+1$ on the edge of the top half $\{y > 0, x^2 + y^2 = 1\}$ of the disk, and at -1 on the edge of the bottom half $\{y < 0, x^2 + y^2 = 1\}$ of the disk.

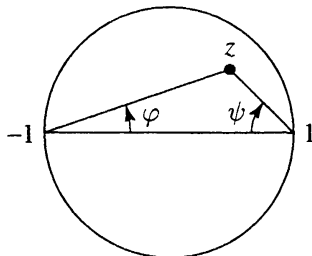
Solution. We recall that the argument function $\text{Arg } w$ has a similar behavior in the upper half-plane $\mathbb{H}$, in that it is constant on each of the intervals $(-\infty, 0)$ and $(0, +\infty)$ of the boundary of $\mathbb{H}$. We map the upper half-plane to the unit disk by a fractional linear transformation so that the interval $(-\infty, 0)$ corresponds to the lower semicircular edge of the disk and $(0, +\infty)$ to the upper semicircular edge. The transformation is given explicitly by

$$z = -\frac{w-i}{w+i}, \quad w = i\frac{1-z}{1+z}.$$

Since $\text{Log } w(z)$ is analytic on the disk, its imaginary part $\text{Arg } w(z)$ is harmonic, and it attains the values 0 on the top edge of the disk and π on the bottom edge. Thus the solution is given by

$$u(z) = 1 - \frac{2}{\pi} \text{Arg } w(z) = \frac{2}{\pi} [\text{Arg}(1+z) - \text{Arg}(1-z)].$$

This solution can be expressed as $u(z) = (2\pi)(\varphi + \psi)$, where φ and ψ are the angles represented in the figure.



Now we turn to electrostatics. Here it is usual to start with the electric force field $\mathbf{E}$ on D , defined so that $\mathbf{E}(A)$ is the force exerted on a unit charge if it were placed at the point A . The work done on a unit charge in moving it along a path γ in D is then given by the integral $\int_{\gamma} \mathbf{E} \cdot \mathbf{t} ds$, where $\mathbf{t}$ is the unit tangent to the curve, and ds is arc length. The field $\mathbf{E}$ is assumed to be irrotational: $\nabla \times \mathbf{E} = \mathbf{0}$. This is equivalent, at least locally, to assuming that $\int_{\gamma} \mathbf{E} \cdot \mathbf{t} ds$ is independent of path. Thus $\mathbf{E} = \nabla \phi$ for some function ϕ , which is called the **electric field potential**. The work done on a unit charge in moving it along a path γ from A to B is then given by

$$\text{work done} = \int_{\gamma} \mathbf{E} \cdot \mathbf{t} ds = \phi(B) - \phi(A).$$

If there are no sources or sinks in D , then $\nabla \cdot \mathbf{E} = 0$, and this is equivalent to Laplace's equation $\Delta \phi = 0$. Thus the electric field potential ϕ is harmonic in D .

Example. By a “line charge” we mean a uniform distribution of charge along a line (infinite straight wire) in three-dimensional space. We assume that the line is perpendicular to the (x, y) -plane. The electric field is then independent of the z -direction, and it is sufficient to describe it in the (x, y) -plane. The electric field corresponding to a line charge through the origin is

$$\mathbf{E} = \frac{1}{r} \mathbf{u}_r,$$

up to a constant multiple depending upon units, where $\mathbf{u}_r$ is the unit vector in the radial direction. The potential function associated with a line charge at 0 is $\phi = \log r$, since $\nabla \phi = \mathbf{E}$. Note that ϕ is harmonic; however, ϕ has no single-valued conjugate harmonic function. The equipotential lines for $\mathbf{E}$ are circles centered at the origin. Through each such circle $\{|z| = r\}$ there is a positive outward flux given by

$$\int_{|z|=r} \mathbf{E} \cdot \mathbf{n} ds = \int_{|z|=r} \mathbf{E} \cdot \mathbf{u}_r ds = \int_0^{2\pi} d\theta = 2\pi.$$

We say that the origin is a **source** for the electric field $\mathbf{E}$. The origin is a **sink** for the electric field $-\mathbf{E}$.

Exercise. Find the electric field $\mathbf{E}$ and the potential function ϕ in a circular ring whose inner edge is grounded ($\phi = 0$) and whose outer edge is conducting ($\phi = \text{constant}$).

Solution. The problem is invariant under rotations, so we try a potential solution that is invariant under rotations, say $\phi = a \log r + b$. The constant values of the potential on the inner and outer circles determine the constants a and b . The electric field is given by $\mathbf{E} = \nabla\phi = (a/r)\mathbf{u}_r$, which is the electric field of a line charge.

As in the case of fluid flow, problems in electrostatics and in heat flow that are effectively two-dimensional and can be analyzed using complex analysis and conjugate harmonic functions are very special. A typical problem in three dimensions involves a vector field $\mathbf{F}$ that satisfies the conditions

$$\begin{aligned}\nabla \times \mathbf{F} &= \mathbf{0} && (\mathbf{F} \text{ is irrotational}), \\ \nabla \cdot \mathbf{F} &= 0 && (\text{no sources or sinks in } D).\end{aligned}$$

Again, the first condition is equivalent to $\int \mathbf{F} \cdot d\mathbf{x}$ being independent of path, and we say that $\mathbf{F}$ is a **conservative field**. In this case $\mathbf{F} = \nabla\phi$ for some potential function ϕ , at least locally. In terms of the potential function ϕ , the second condition becomes $\Delta\phi = 0$, and ϕ is harmonic. Thus harmonic functions play an important role, as do the Dirichlet and Neumann problems. However, in three-dimensional problems the harmonic functions cannot be analyzed as the real parts of analytic functions, and conformal mapping techniques are not available.

Exercises for III.7

1. Find the steady-state heat distribution $u(x, y)$ in a laminar plate corresponding to the half-disk $\{x^2 + y^2 < 1, y > 0\}$, where the semicircular top edge is held at temperature T_1 and the lower edge $(-1, 1)$ is held at temperature T_2 . Find and sketch the isothermal curves for the heat distribution. *Hint.* Consider the steady-state heat distribution for the full unit disk with the top held at temperature T_1 and the bottom at temperature T_3 , where $T_2 = (T_1 + T_3)/2$.
2. Find the potential function $\phi(x, y)$ for the electric field for a conducting laminar plate corresponding to the quarter-disk $\{x^2 + y^2 < 1, x > 0, y > 0\}$, where the two edges on the coordinate axes are grounded (that is, $\phi = 0$ on the edges), and the semicircular edge is held at constant potential V_1 . Find and sketch the equipotential lines and the lines of force for the electric field. *Hint.* Use the conformal map $\zeta = z^2$ and the solution to the preceding exercise.
3. Find the potential function $\phi(x, y)$ for the electric field for a conducting laminar plate corresponding to the unit disk where the boundary

quarter-circles in each quadrant are held at constant voltages V_1 , V_2 , V_3 , and V_4 . *Hint.* Map the disk to the upper half-plane by $w = w(z)$ and consider potential functions of the form $\text{Arg}(w - a)$.

4. Find the steady-state heat distribution in a laminar plate corresponding to the vertical half-strip $\{|x| < \pi/2, y > 0\}$, where the vertical sides at $x = \pm\pi/2$ are held at temperature $T_0 = 0$ and the bottom edge $(-\pi/2, \pi/2)$ on the real axis is held at temperature $T_1 = 100$. Make a rough sketch of the isothermal curves and the lines of heat flow. *Hint.* Use $w = \sin z$ to map the strip to the upper half-plane, and make use of harmonic functions of the form $\text{Arg}(w - a)$.
5. Find the steady-state heat distribution in a laminar plate corresponding to the vertical half-strip $\{|x| < \pi/2, y > 0\}$, where the side $x = -\pi/2$ is held at constant temperature T_0 , the side $x = \pi/2$ is held at constant temperature T_1 , and the bottom edge $(-\pi/2, \pi/2)$ on the real axis is insulated; that is, no heat passes through the bottom edge, so the gradient ∇u of the solution $u(x, y)$ is parallel there to the x -axis. *Hint.* Try linear functions plus constants.
6. Find the steady-state heat distribution in a laminar plate corresponding to the upper half-plane $\{y > 0\}$, where the interval $(-1, 1)$ is insulated, the interval $(-\infty, -1)$ is held at temperature T_0 , and the interval $(1, \infty)$ is held at temperature T_1 . Make a rough sketch of the isothermal curves and the lines of heat flow. *Hint.* Use the solution to the preceding exercise and an appropriate conformal map.
7. The gravitational field near the surface of the earth is approximately constant, of the form $\mathbf{F} = c\mathbf{k}$, where $\mathbf{k}$ is the unit vector in the z -direction in (x, y, z) -space and the surface of the earth is represented by the plane where $z = 0$ (the flat earth theory). Show that $\mathbf{F}$ is conservative, and find a potential function ϕ for $\mathbf{F}$.
8. Show that the inverse square force field $\mathbf{F} = \mathbf{u}_r/r^2$ on $\mathbb{R}^3$ is conservative. Find the potential function ϕ for $\mathbf{F}$, and show that ϕ is harmonic.
9. For $n \geq 3$, show that the function $1/r^{n-2}$ is harmonic on $\mathbb{R}^n \setminus \{0\}$. Find the vector field $\mathbf{F}$ that has this function as its potential.

IV

Complex Integration and Analyticity

In this chapter we take up the complex integral calculus. In Section 1 we introduce complex line integrals, and in Section 2 we develop the complex integral calculus, emphasizing the analogy with the usual one-variable integral calculus. In Section 3 we lay the cornerstone of the complex integral calculus, which is Cauchy's theorem. The version we prove is an immediate consequence of Green's theorem. In Section 4 we derive the Cauchy integral formula and use it to show that analytic functions have analytic derivatives. Each of the final four sections features a "named" theorem. In Section 5 we prove Liouville's theorem. In Section 6 we give a version of Morera's theorem that provides a useful criterion for determining whether a continuous function is analytic. Sections 7 and 8, on Goursat's theorem and the Pompeiu formula, can be omitted at first reading.

1. Complex Line Integrals

For complex analysis it is convenient to define $dz = dx + i dy$. According to this notation, if $h(z)$ is a complex-valued function on a curve γ , then

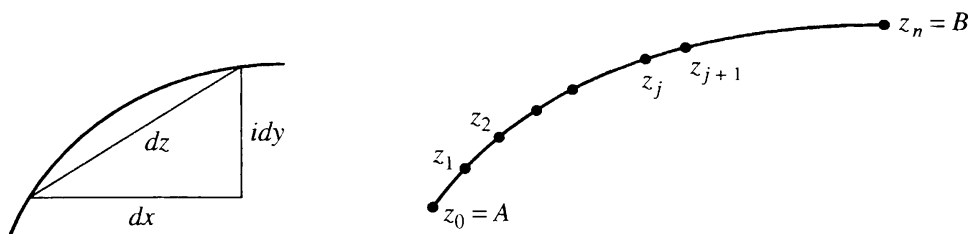
$$(1.1) \quad \int_{\gamma} h(z) dz = \int_{\gamma} h(z) dx + i \int_{\gamma} h(z) dy.$$

Suppose γ is parameterized by $t \mapsto z(t) = x(t) + iy(t)$, $a \leq t \leq b$. The Riemann sum approximating $\int_{\gamma} h(z)(dx + i dy)$ corresponding to the subdivision $a = t_0 < t_1 < \cdots < t_n = b$ is given by

$$\sum h(z_j)(x_{j+1} - x_j) + i \sum h(z_j)(y_{j+1} - y_j),$$

where $z(t_j) = z_j = x_j + iy_j$. If we express these sums in terms of the z_j 's, we obtain the Riemann sum approximation

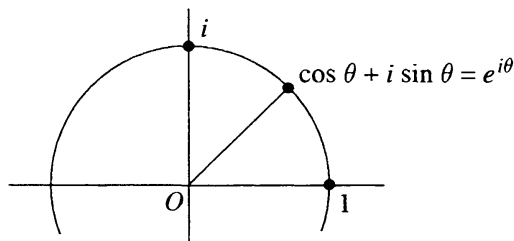
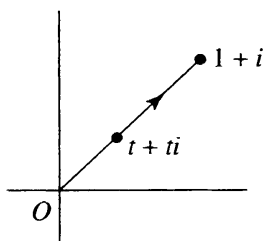
$$(1.2) \quad \int_{\gamma} h(z) dz \approx \sum h(z_j)(z_{j+1} - z_j).$$



This expression justifies the notation dz .

Example. To compute $\int_0^{1+i} z^2 dz$ along the straight line segment from 0 to $1+i$, we parametrize the line segment by $z(t) = t + ti$, $0 \leq t \leq 1$, so that $x(t) = t$ and $y(t) = t$. Then $dz = dx + i dy = (1+i)dt$, and we obtain

$$\int_0^{1+i} z^2 dz = \int_0^1 [(1+i)t]^2 (1+i) dt = (1+i)^3 \int_0^1 t^2 dt = \frac{(1+i)^3}{3}.$$



Example. We evaluate

$$\oint_{|z|=1} \frac{dz}{z},$$

where we integrate around the unit circle in the usual positive (counter-clockwise) direction. The unit circle is parameterized by $z(\theta) = e^{i\theta} = \cos \theta + i \sin \theta$, $0 \leq \theta \leq 2\pi$, so that $dx = -\sin \theta d\theta$, $dy = \cos \theta d\theta$, and

$$dz = dx + i dy = -\sin \theta d\theta + i \cos \theta d\theta = i(\cos \theta + i \sin \theta) d\theta.$$

Thus

$$\frac{dz}{z} = i d\theta,$$

and the integral becomes

$$\oint_{|z|=1} \frac{dz}{z} = i \int_0^{2\pi} d\theta = 2\pi i.$$

Note in this calculation that $dz = ie^{i\theta} d\theta$, which is what is obtained by applying the usual rule for differentiation of exponentials to $z(\theta) = e^{i\theta}$.

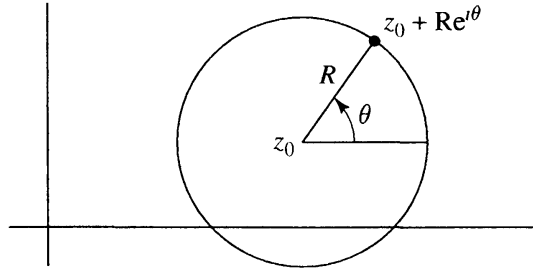
Example. For m an integer and $R > 0$, we show that

$$\oint_{|z-z_0|=R} (z-z_0)^m dz = \begin{cases} 0, & m \neq -1, \\ 2\pi i, & m = -1. \end{cases}$$

We parametrize the circle $|z - z_0| = R$, with the usual positive orientation, by $z(\theta) = z_0 + Re^{i\theta}$, and we calculate that $dz = iRe^{i\theta}d\theta$. Thus the integral becomes

$$\int_0^{2\pi} (Re^{i\theta})^m iRe^{i\theta} d\theta = iR^{m+1} \int_0^{2\pi} e^{i(m+1)\theta} d\theta.$$

The integral on the right is 0 unless $m = -1$, in which case it is 2π . This yields the formula.



In complex analysis it is customary to denote the infinitesimal arc length d by $|dz|$:

$$|dz| = ds = \sqrt{(dx)^2 + (dy)^2}.$$

This means that if a curve γ is parameterized by $z(t) = x(t) + iy(t)$, then

$$(1.3) \quad \int_{\gamma} h(z)|dz| = \int_{\gamma} h(z) ds = \int_a^b h(z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

In particular, the length of γ is

$$L = \int_{\gamma} |dz| = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

The notation can be justified by considering the sums approximating these integrals. For the subdivision of the parameter interval used earlier, the usual sum used in multivariable calculus to approximate $\int_{\gamma} h(x, y) ds$ is

$$\int_{\gamma} h(x, y) ds \approx \sum h(x_j, y_j) \sqrt{(x_{j+1} - x_j)^2 + (y_{j+1} - y_j)^2}.$$

In complex notation this becomes

$$(1.4) \quad \int_{\gamma} h(z) |dz| \approx \sum h(z_j) |z_{j+1} - z_j|.$$

In particular, the sum approximating the length L of γ is given by

$$(1.5) \quad L \approx \sum |z_{j+1} - z_j|.$$

Example. The parameterization $z(\theta) = z_0 + Re^{i\theta}$ of the circle $|z - z_0| = R$ can be used to derive the expression for the infinitesimal arc length for

the circle in terms of the central angle θ . In this case, $x(\theta) = R \cos \theta$, so that $dx = -R \sin \theta d\theta$, and similarly $dy = R \cos \theta d\theta$. Thus $|dz| = ds = \sqrt{(dx)^2 + (dy)^2} = R d\theta$. The length of the circle is $\int_0^{2\pi} |dz| = 2\pi R$, as usual.

Theorem. Suppose γ is a piecewise smooth curve. If $h(z)$ is a continuous function on γ , then

$$(1.6) \quad \left| \int_{\gamma} h(z) dz \right| \leq \int_{\gamma} |h(z)| |dz|.$$

Further, if γ has length L , and $|h(z)| \leq M$ on γ , then

$$(1.7) \quad \left| \int_{\gamma} h(z) dz \right| \leq ML.$$

The estimate (1.6) is the triangle inequality for integrals. It will be used frequently and without reference. The estimate (1.7) is called the *ML-estimate*. It may or may not be referred to. These estimates both follow from the corresponding estimates for the approximating sums. The triangle inequality (1.6) for integrals follows from the triangle inequality for the approximating Riemann sums,

$$\left| \sum h(z_j)(z_{j+1} - z_j) \right| \leq \sum |h(z_j)| |z_{j+1} - z_j|,$$

in view of (1.2) and (1.4). The *ML-estimate* (1.7) follows from the estimate

$$\left| \sum h(z_j)(z_{j+1} - z_j) \right| \leq M \sum |z_{j+1} - z_j|,$$

in view of (1.2) and (1.5).

Example. If we apply the *ML-estimate* to the first example above, we obtain

$$\left| \int_0^{1+i} z^2 dz \right| \leq 2\sqrt{2},$$

since $|z^2| \leq |1+i|^2 = 2$ on the straight line segment from 0 to $1+i$, and the segment has length $L = \sqrt{2}$. Since the value of the integral has modulus $|1+i|^3/3 = 2\sqrt{2}/3$, in this case the *ML-estimate* provides only a rough estimate. A better estimate is obtained by noting that $|dz| = \sqrt{2}dt$ on the line and applying (1.6):

$$\left| \int_0^{1+i} z^2 dz \right| \leq \int_0^{1+i} |z^2| |dz| = \int_0^1 (\sqrt{2}t)^2 \sqrt{2} dt = \frac{2\sqrt{2}}{3}.$$

Since equality actually holds here, this estimate cannot be improved. We say that the estimate is a **sharp estimate**.

Example. We apply the ML -estimate to

$$\oint_{|z-z_0|=R} \frac{1}{z-z_0} dz = 2\pi i.$$

In this case the integrand has constant modulus $1/R$ on the circle of integration, so we take $M = 1/R$ and $L = 2\pi R$. The ML -estimate becomes

$$\left| \oint_{|z-z_0|=R} \frac{1}{z-z_0} dz \right| \leq 2\pi.$$

Since equality actually holds here, the ML -estimate is sharp.

Exercises for IV.1

- Let γ be the boundary of the triangle $\{0 < y < 1-x, 0 < x < 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals.

$$(a) \int_{\gamma} \operatorname{Re} z \, dz \quad (b) \int_{\gamma} \operatorname{Im} z \, dz \quad (c) \int_{\gamma} z \, dz$$

- Let γ be the unit circle $\{|z| = 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \dots$

$$(a) \int_{\gamma} z^m \, dz \quad (b) \int_{\gamma} \bar{z}^m \, dz \quad (c) \int_{\gamma} z^m |dz|$$

- Let γ be the circle $\{|z| = R\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \dots$

$$(a) \int_{\gamma} |z^m| \, dz \quad (b) \int_{\gamma} |z^m| |dz| \quad (c) \int_{\gamma} \bar{z}^m \, dz$$

- Show that if D is a bounded domain with smooth boundary, then

$$\int_{\partial D} \bar{z} \, dz = 2i \operatorname{Area}(D).$$

- Show that

$$\left| \oint_{|z-1|=1} \frac{e^z}{z+3} dz \right| \leq 2\pi e^2.$$

- Show that

$$\left| \oint_{|z|=R} \frac{\operatorname{Log} z}{z^2} dz \right| \leq 2\sqrt{2}\pi \frac{\log R}{R}, \quad R > e^{\pi}.$$

- Show that there is a strict inequality

$$\left| \oint_{|z|=R} \frac{z^n}{z^m - 1} dz \right| < \frac{2\pi R^{n+1}}{R^m - 1}, \quad R > 1, \quad m \geq 1, \quad n \geq 0.$$

8. Suppose the continuous function $f(e^{i\theta})$ on the unit circle satisfies $|f(e^{i\theta})| \leq M$ and $|\int_{|z|=1} f(z) dz| = 2\pi M$. Show that $f(z) = c\bar{z}$ for some constant c with modulus $|c| = M$.
9. Suppose $h(z)$ is a continuous function on a curve γ . Show that

$$H(w) = \int_{\gamma} \frac{h(z)}{z - w} dz, \quad w \in \mathbb{C} \setminus \gamma,$$

is analytic on the complement of γ , and find $H'(w)$.

2. Fundamental Theorem of Calculus for Analytic Functions

Let $f(z)$ be a continuous function on a domain D . A function $F(z)$ on D is a **(complex) primitive** for $f(z)$ if $F(z)$ is analytic and $F'(z) = f(z)$. The following theorem is the analogue of the first statement of the fundamental theorem of calculus.

Theorem (Part I). *If $f(z)$ is continuous on a domain D , and if $F(z)$ is a primitive for $f(z)$, then*

$$\int_A^B f(z) dz = F(B) - F(A),$$

where the integral can be taken over any path in D from A to B .

This formula follows from the corresponding formula for line integrals. In this case, we have

$$F'(z) = \frac{\partial F}{\partial x} = \frac{1}{i} \frac{\partial F}{\partial y},$$

so that

$$\begin{aligned} F(B) - F(A) &= \int_A^B dF = \int_A^B \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \\ &= \int_A^B F'(z) (dx + i dy) = \int_A^B F'(z) dz. \end{aligned}$$

This theorem provides a powerful tool for evaluating definite integrals. The problem of evaluating $\int_A^B f(z) dz$ is reduced to that of finding an analytic function $F(z)$ whose derivative is $f(z)$.

Example. To integrate z^2 from 1 to i , we observe that $z^3/3$ is a primitive for z^2 , and then we proceed as we would to evaluate an ordinary garden-variety integral,

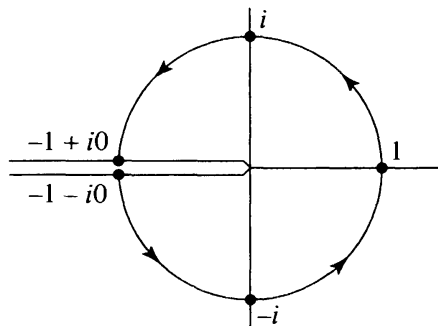
$$\int_0^{1+i} z^2 dz = \left. \frac{z^3}{3} \right|_0^{1+i} = \frac{(1+i)^3}{3}.$$

This coincides with the result that was obtained in the preceding section by parameterizing the straight line from 0 to $1 + i$. Note, though, that the integral is independent of path.

Example. The function $1/z$ does not have an analytic primitive defined on any domain containing the unit circle. To evaluate the integral of dz/z around the unit circle we can still use the primitive $F(z) = \text{Log } z$ in the slit plane $\mathbb{C} \setminus (-\infty, 0]$, by taking the integral over a path counterclockwise around the unit circle starting at a point $-1 - 0i$ just below the slit and ending at a point $-1 + 0i$ just above the slit. Then

$$\int_{|z|=1} \frac{dz}{z} = \text{Log } z \Big|_{-1-0i}^{-1+0i} = i\pi - i(-\pi) = 2\pi i,$$

which coincides with the result obtained earlier by parameterizing the unit circle.



The analogue of the second statement in the fundamental theorem of calculus is that every analytic function on a star-shaped domain has a primitive, which can be defined as an indefinite complex integral.

Theorem (Part II). *Let D be a star-shaped domain, and let $f(z)$ be analytic on D . Then $f(z)$ has a primitive on D , and the primitive is unique up to adding a constant. A primitive for $f(z)$ is given explicitly by*

$$F(z) = \int_{z_0}^z f(\zeta) d\zeta, \quad z \in D,$$

where z_0 is any fixed point of D , and where the integral can be taken along any path in D from z_0 to z .

To see this, we write $f = u + iv$ as usual, and we consider the differential $u dx - v dy$. Since f is analytic, the Cauchy-Riemann equations yield $\partial u / \partial y = -\partial v / \partial x$, and the differential is closed. By the fundamental theorem of calculus for line integrals (Section III.2), the differential is exact on D , that is, there is a continuously differentiable function U on D such

that $dU = u dx - v dy$. In other words, U satisfies

$$\frac{\partial U}{\partial x} = u, \quad \frac{\partial U}{\partial y} = -v.$$

These equations show that U is in fact twice continuously differentiable. Another application of the Cauchy-Riemann equations yields

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0.$$

Hence U is harmonic. Since D is star-shaped, there is a conjugate harmonic function V for U on D , so that $G = U + iV$ is analytic on D . Then

$$G' = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x} = \frac{\partial U}{\partial x} - i \frac{\partial U}{\partial y} = u + iv = f,$$

and $G(z)$ is a primitive for $f(z)$. If $G_0(z)$ is another primitive for $f(z)$, then $G - G_0$ has derivative zero, and $G - G_0$ is constant on D . If we apply Part I above to the primitive $F(z) = G(z) - G(z_0)$ for $f(z)$, we obtain the formula in Part II.

Note in particular that integrals of analytic functions in star-shaped domains are independent of path. This is not true for arbitrary domains. The identity

$$\oint_{|z|=1} \frac{dz}{z} = 2\pi i \neq 0$$

shows that the differential dz/z does not have a primitive on the punctured plane $\mathbb{C} \setminus \{0\}$.

Exercises for IV.2

- Evaluate the following integrals, for a path γ that travels from $-\pi i$ to πi in the right half-plane, and also for a path γ from $-\pi i$ to πi in the left half-plane.
 - $\int_{\gamma} z^4 dz$
 - $\int_{\gamma} e^z dz$
 - $\int_{\gamma} \cos z dz$
 - $\int_{\gamma} \sinh z dz$
- Using an appropriate primitive, evaluate $\int_{\gamma} 1/z dz$ for a path γ that travels from $-\pi i$ to πi in the right half-plane, and also for a path γ from $-\pi i$ to πi in the left half-plane. For each path give a precise definition of the primitive used to evaluate the integral.
- Show that if $m \neq -1$, then z^m has a primitive on $\mathbb{C} \setminus \{0\}$.
- Let $D = \mathbb{C} \setminus (-\infty, 1]$, and consider the branch of $\sqrt{z^2 - 1}$ on D that is positive on the interval $(1, \infty)$. (a) Show that $z + \sqrt{z^2 - 1}$ omits the negative real axis, that is, the range of the function on D does not include any values in the interval $(-\infty, 0]$ on the real axis. (b)

Show that $\text{Log}(z + \sqrt{z^2 - 1})$ is a primitive for $1/\sqrt{z^2 - 1}$ on D . (c) Evaluate

$$\int_{\gamma} \frac{dz}{\sqrt{z^2 - 1}},$$

where γ is the path from $-2i$ to $+2i$ in D counterclockwise around the circle $|z| = 2$. (d) Evaluate the integral above in the case γ is the entire circle $|z| = 2$, oriented counterclockwise. (Note that the primitive is discontinuous at $z = -2$.)

5. Show that an analytic function $f(z)$ has a primitive in D if and only if $\int_{\gamma} f(z)dz = 0$ for every closed path γ in D .

3. Cauchy's Theorem

We begin with a smooth complex-valued function $f(z) = u + iv$, and we express

$$f(z)dz = (u + iv)(dx + i dy) = (u + iv)dx + (-v + iu)dy.$$

The condition that $f(z)dz$ be a closed differential is

$$\frac{\partial}{\partial y}(u + iv) = \frac{\partial}{\partial x}(-v + iu).$$

Taking the real and imaginary parts, we see that this is equivalent to

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \frac{\partial v}{\partial y} = \frac{\partial u}{\partial x},$$

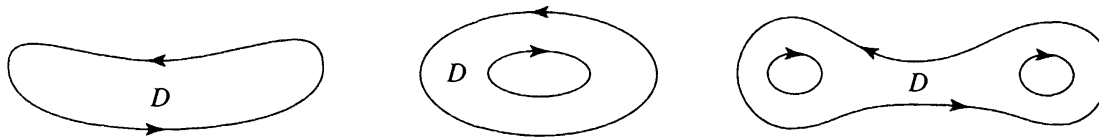
which are the Cauchy-Riemann equations for u and v . Thus we obtain the following theorem, which is the original form of Morera's theorem.

Theorem. *A continuously differentiable function $f(z)$ on D is analytic if and only if the differential $f(z)dz$ is closed.*

From Green's theorem (Section III.1.1) we obtain immediately the following far-reaching theorem.

Theorem (Cauchy's Theorem). *Let D be a bounded domain with piecewise smooth boundary. If $f(z)$ is an analytic function on D that extends smoothly to ∂D , then*

$$\int_{\partial D} f(z)dz = 0.$$



Example. If we apply Cauchy's theorem to a function $f(z)$ analytic on the annulus $D = \{r < |z| < R\}$, we obtain

$$0 = \int_{\partial D} f(z) dz = \oint_{|z|=R} f(z) dz - \oint_{|z|=r} f(z) dz.$$

Note the change in sign, due to the fact that the inside boundary circle is traversed in the negative (clockwise) direction according to its orientation as a boundary curve for D . Thus

$$\oint_{|z|=R} f(z) dz = \oint_{|z|=r} f(z) dz.$$

This is in accord with the discussion in Section III.2, since the differential $f(z)dz$ is closed, and since the outside circle of the annulus can be continuously deformed to the inside circle of the annulus through a family of closed curves (intermediate circles).

Exercises for IV.3

1. By integrating $e^{-z^2/2}$ around a rectangle with vertices $\pm R$, $it \pm R$, and sending R to ∞ , show that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} e^{-itx} dx = e^{-t^2/2}, \quad -\infty < t < \infty.$$

Use the known value of the integral for $t = 0$. *Remark.* This shows that $e^{-x^2/2}$ is an eigenfunction of the Fourier transform with eigenvalue 1. For more, see the next exercise.

2. We define the **Hermite polynomials** $H_n(x)$ and **Hermite orthogonal functions** $\phi_n(x)$ for $n \geq 0$ by

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}), \quad \phi_n(x) = e^{-x^2/2} H_n(x).$$

- (a) Show that $H_n(x) = 2^n x^n + \dots$ is a polynomial of degree n that is even when n is even and odd when n is odd.
- (b) By integrating the function

$$e^{(z-it)^2/2} \frac{d^n}{dz^n} (e^{-z^2})$$

around a rectangle with vertices $\pm R$, $it \pm R$ and sending R to ∞ , show that

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi_n(x) e^{-itx} dx = (-i)^n \phi_n(t), \quad -\infty < t < \infty.$$

Hint. Use the identity from Exercise 1, and also justify and use the identity

$$\frac{d^n}{dx^n} e^{-(x+it)^2} = \frac{1}{i^n} \frac{d^n}{dt^n} e^{-(x+it)^2}.$$

- (c) Show that $\phi_n'' - x^2\phi_n + (2n+1)\phi_n = 0$.
 (d) Using $\int \phi_n'' \phi_m dx = \int \phi_n \phi_m'' dx$ and (c), show that

$$\int_{-\infty}^{\infty} \phi_n(x) \phi_m(x) dx = 0, \quad n \neq m.$$

Remark. This shows that the ϕ_n 's form an orthogonal system of eigenfunctions for the (normalized) Fourier transform operator $\mathcal{F}$ with eigenvalues ± 1 and $\pm i$. Thus $\mathcal{F}$ extends to a unitary operator on square-integrable functions. Further, $\mathcal{F}^4$ is the identity operator, and the inverse Fourier transform is given by $(\mathcal{F}^{-1}f)(x) = (\mathcal{F}f)(-x)$.

3. Let $f(z) = c_0 + c_1z + \cdots + c_nz^n$ be a polynomial.
 (a) If the c_k 's are real, show that

$$\int_{-1}^1 f(x)^2 dx \leq \pi \int_0^{2\pi} |f(e^{i\theta})|^2 \frac{d\theta}{2\pi} = \pi \sum_{k=0}^n c_k^2.$$

Hint. For the first inequality, apply Cauchy's theorem to the function $f(z)^2$ separately on the top half and the bottom half of the unit disk.

- (b) If the c_k 's are complex, show that

$$\int_{-1}^1 |f(x)|^2 dx \leq \pi \int_0^{2\pi} |f(e^{i\theta})|^2 \frac{d\theta}{2\pi} = \pi \sum_{k=0}^n |c_k|^2.$$

- (c) Establish the following variant of **Hilbert's inequality**, that

$$\left| \sum_{j,k=0}^n \frac{c_j c_k}{j+k+1} \right| \leq \pi \sum_{k=0}^n |c_k|^2,$$

with strict inequality unless the complex numbers $c_0, \dots, c_n$ are all zero. *Hint.* Start by evaluating $\int_0^1 f(x)^2 dx$.

4. Prove that a polynomial in z without zeros is constant (the fundamental theorem of algebra) using Cauchy's theorem, along the following lines. If $P(z)$ is a polynomial that is not a constant, write $P(z) = P(0) + zQ(z)$, divide by $zP(z)$, and integrate around a large circle. This will lead to a contradiction if $P(z)$ has no zeros.

5. Suppose that D is a bounded domain with piecewise smooth boundary, and that $f(z)$ is analytic on $D \cup \partial D$. Show that

$$\sup_{z \in \partial D} |\bar{z} - f(z)| \geq 2 \frac{\text{Area}(D)}{\text{Length}(\partial D)}.$$

Show that this estimate is sharp, and that in fact there exist D and $f(z)$ for which equality holds. *Hint.* Consider $\int_{\partial D} [\bar{z} - f(z)] dz$, and use Exercise 4 in Section 1.

6. Suppose $f(z)$ is continuous in the closed disk $\{|z| \leq R\}$ and analytic on the open disk $\{|z| < R\}$. Show that $\oint_{|z|=R} f(z) dz = 0$. *Hint.* Approximate $f(z)$ uniformly by $f_r(z) = f(rz)$.

4. The Cauchy Integral Formula

Integral representation formulae are powerful tools for studying functions. One application of an integral representation is to estimate the size of the function being represented. Another is to obtain formulae for derivatives, by differentiating under the integral sign. The prototype of the integral representation is provided by the Cauchy integral formula, representing an analytic function. The integral representation will allow us to show that all the derivatives of an analytic function are analytic. It will also allow us to obtain power series expansions for analytic functions.

Theorem (Cauchy Integral Formula). *Let D be a bounded domain with piecewise smooth boundary. If $f(z)$ is analytic on D , and $f(z)$ extends smoothly to the boundary of D , then*

$$(4.1) \quad f(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - z} dw, \quad z \in D.$$

To establish the formula, fix a point $z \in D$, let $\varepsilon > 0$ be small, and consider the domain $D_\varepsilon = D \setminus \{|w - z| \leq \varepsilon\}$ obtained from D by punching out a disk centered at z of radius ε . The boundary ∂D_ε is the union of ∂D and the circle $\{|w - z| = \varepsilon\}$, oriented clockwise. Since $f(z)/(w - z)$ is analytic for $w \in D_\varepsilon$, Cauchy's theorem yields

$$\int_{\partial D_\varepsilon} \frac{f(w)}{w - z} dw = 0.$$

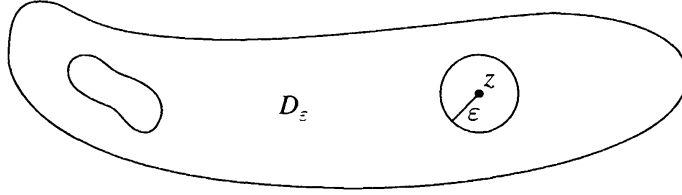
Reversing the orientation of the circle to counterclockwise produces a sign change, and we obtain

$$\int_{|w-z|=\varepsilon} \frac{f(w)}{w - z} dw = \int_{\partial D} \frac{f(w)}{w - z} dw.$$

Writing $w = z + \varepsilon e^{i\theta}$, $dw = i\varepsilon e^{i\theta} d\theta$, and dividing by $2\pi i$, we obtain

$$\int_0^{2\pi} f(z + \varepsilon e^{i\theta}) \frac{d\theta}{2\pi} = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - z} dw.$$

By the mean value property of harmonic functions, the integral on the left-hand side coincides with $f(z)$, and the formula is established.



We remark that without the mean value property at our disposal, we could complete the argument by observing that the integral on the left is the average value of $f(w)$ on the circle centered at z of radius ε . Since $f(w)$ is continuous at z , these averages tend to $f(z)$ as $\varepsilon \rightarrow 0$. This can be justified with complete rigor by writing

$$\int_0^{2\pi} f(z + \varepsilon e^{i\theta}) \frac{d\theta}{2\pi} = f(z) + \int_0^{2\pi} [f(z + \varepsilon e^{i\theta}) - f(z)] \frac{d\theta}{2\pi},$$

and showing by a direct estimate that the latter integral tends to 0 as $\varepsilon \rightarrow 0$.

If we differentiate under the integral sign and use

$$\frac{d^m}{dz^m} \frac{1}{w - z} = \frac{m!}{(w - z)^{m+1}},$$

we obtain integral formulae for the derivatives $f^{(m)}(z)$ of $f(z)$.

Theorem. *Let D be a bounded domain with piecewise smooth boundary. If $f(z)$ is an analytic function on D that extends smoothly to the boundary of D , then $f(z)$ has complex derivatives of all orders on D , which are given by*

$$(4.2) \quad f^{(m)}(z) = \frac{m!}{2\pi i} \int_{\partial D} \frac{f(w)}{(w - z)^{m+1}} dw, \quad z \in D, \quad m \geq 0.$$

The case $m = 0$ of (4.2) is the Cauchy integral formula (4.1). Though it is not really necessary, we give separately the argument for the case $m = 1$. This special case already includes all the ideas for the proof of the general case.

Using (4.1) and the identity

$$\frac{1}{w - (z + \Delta z)} - \frac{1}{w - z} = \frac{\Delta z}{(w - (z + \Delta z))(w - z)},$$

we express the difference quotient approximating $f'(z)$ as

$$\begin{aligned}\frac{f(z + \Delta z) - f(z)}{\Delta z} &= \frac{1}{\Delta z} \left[\frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - (z + \Delta z)} dw - \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{w - z} dw \right] \\ &= \frac{1}{2\pi i} \int_{\partial D} f(w) \frac{1}{(w - (z + \Delta z))(w - z)} dw.\end{aligned}$$

As Δz tends to 0, the integrand converges to $f(w)/(w - z)^2$, uniformly for $w \in \partial D$. Hence the integrals converge, and we obtain in the limit that

$$f'(z) = \frac{1}{2\pi i} \int_{\partial D} \frac{f(w)}{(w - z)^2} dw, \quad z \in D,$$

which is the case $m = 1$ of (4.2).

The general case of (4.2) is proved by induction on m . We assume that $f^{(m-2)}(z)$ is complex differentiable and the formula holds for $f^{(m-1)}(z)$. We must show that $f^{(m-1)}(z)$ is complex differentiable and the formula holds for $f^{(m)}(z)$. Using the binomial expansion

$$(w - z - \Delta z)^m = (w - z)^m - m(w - z)^{m-1}\Delta z + \frac{m(m-1)}{2}(w - z)^{m-2}(\Delta z)^2 + \dots$$

and simplifying, we obtain

$$\begin{aligned}\frac{1}{(w - (z + \Delta z))^m} - \frac{1}{(w - z)^m} &= \frac{(w - z)^m - (w - z - \Delta z)^m}{(w - z)^m(w - z - \Delta z)^m} \\ &= \frac{m\Delta z}{(w - z)(w - z - \Delta z)^m} - \frac{m(m-1)(\Delta z)^2}{2(w - z)^2(w - z - \Delta z)^m} + \dots,\end{aligned}$$

where the dots indicate terms with powers of Δz up to $(\Delta z)^m$. The integral formula for $f^{(m-1)}(z)$ then yields the expression

$$\frac{(m-1)!}{2\pi i} \int_{\partial D} f(w) \left[\frac{m}{(w - z)(w - z - \Delta z)^m} + \Delta z(\dots) \right] dw$$

for the difference quotient $(f^{(m-1)}(z + \Delta z) - f^{(m-1)}(z))/\Delta z$. Again the integrand converges as $\Delta z \rightarrow 0$, uniformly for $w \in \partial D$, and we can pass to the limit to conclude that $f^{(m)}(z)$ exists and is given by (4.2).

Since each of the successive complex derivatives of $f(z)$ is complex differentiable, each is continuous, and thus each is analytic. The hypothesis that the domain has smooth boundary is irrelevant for determining analyticity, as we can restrict the function to an appropriate small disk. Thus we have proved the following.

Corollary. *If $f(z)$ is analytic on a domain D , then $f(z)$ is infinitely differentiable, and the successive complex derivatives $f'(z)$, $f''(z)$, $\dots$, are all analytic on D .*

Example. The Cauchy integral formula for z^2 yields

$$\oint_{|z|=2} \frac{z^2}{z-1} dz = 2\pi i z^2 \Big|_{z=1} = 2\pi i.$$

Example. The Cauchy integral formula for the derivative of $z^2 \sin z$ yields

$$\oint_{|z|=2\pi} \frac{z^2 \sin z}{(z-\pi)^3} dz = \frac{2\pi i}{2!} \frac{d^2}{dz^2} (z^2 \sin z) \Big|_{z=\pi} = \pi i (4\pi \cos \pi) = -4\pi^2 i.$$

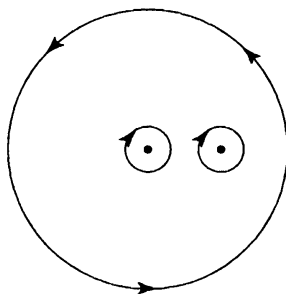
Example. Consider the integral

$$\oint_{|z|=5} \frac{e^z}{z^2(z-1)} dz,$$

which does not have the form given in the Cauchy integral formula. By applying Cauchy's theorem to the domain D_ε obtained by excising two small disks centered at 0 and 1 from the disk $\{|z| < 2\}$, we can express the integral above as the sum of two integrals, each of which can be evaluated by the Cauchy formula:

$$\begin{aligned} \oint_{|z|=2} \frac{e^z}{z^2(z-1)} dz &= \oint_{|z|=\varepsilon} \frac{e^z}{z^2(z-1)} dz + \oint_{|z-1|=\varepsilon} \frac{e^z}{z^2(z-1)} dz \\ &= 2\pi i \frac{d}{dz} \frac{e^z}{(z-1)} \Big|_{z=0} + 2\pi i \frac{e^z}{z^2} \Big|_{z=1} \\ &= -2\pi i - 2\pi i + 2\pi i e = 2\pi i(e-2). \end{aligned}$$

These integrals can also be handled by residue theory, as we shall soon see.



Exercises for IV.4

1. Evaluate the following integrals, using the Cauchy integral formula:

- | | |
|--|--|
| (a) $\oint_{ z =2} \frac{z^n}{z-1} dz, \quad n \geq 0$ | (e) $\oint_{ z =1} \frac{e^z}{z^m} dz, \quad -\infty < m < \infty$ |
| (b) $\oint_{ z =1} \frac{z^n}{z-2} dz, \quad n \geq 0$ | (f) $\int_{ z-1-i =5/4} \frac{\text{Log } z}{(z-1)^2} dz$ |
| (c) $\oint_{ z =1} \frac{\sin z}{z} dz$ | (g) $\oint_{ z =1} \frac{dz}{z^2(z^2-4)e^z}$ |

$$(d) \oint_{|z|=1} \frac{\cosh z}{z^3} dz \qquad (h) \oint_{|z-1|=3} \frac{dz}{z(z^2-4)e^z}$$

2. Show that a harmonic function is C^∞ , that is, a harmonic function has partial derivatives of all orders.
3. Use the Cauchy integral formula to derive the mean value property of harmonic functions, that

$$u(z_0) = \int_0^{2\pi} u(z_0 + \rho e^{i\theta}) \frac{d\theta}{2\pi}, \quad z_0 \in D,$$

whenever $u(z)$ is harmonic in a domain D and the closed disk $|z - z_0| \leq \rho$ is contained in D .

4. Let D be a bounded domain with smooth boundary ∂D , and let $z_0 \in D$. Using the Cauchy integral formula, show that there is a constant C such that

$$|f(z_0)| \leq C \sup\{|f(z)| : z \in \partial D\}$$

for any function $f(z)$ analytic on $D \cup \partial D$. By applying this estimate to $f(z)^n$, taking n th roots, and letting $n \rightarrow \infty$, show that the estimate holds with $C = 1$. *Remark.* This provides an alternative proof of the maximum principle for analytic functions.

5. Liouville's Theorem

Suppose that $f(z)$ is analytic on the closed disk $\{|z - z_0| \leq \rho\}$, that is, it is analytic on some domain containing the closed disk. By the Cauchy integral formula for $f^{(m)}(z)$,

$$f^{(m)}(z_0) = \frac{m!}{2\pi i} \int_{|z-z_0|=\rho} \frac{f(z)}{(z-z_0)^{m+1}} dz.$$

We parametrize the boundary circle by $z = z_0 + \rho e^{i\theta}$, $dz = i\rho e^{i\theta} d\theta$. Then

$$\frac{1}{2\pi i} \frac{f(z)}{(z-z_0)^{m+1}} dz = \frac{f(z_0 + \rho e^{i\theta})}{\rho^m e^{im\theta}} \frac{d\theta}{2\pi},$$

and we obtain

$$f^{(m)}(z_0) = \frac{m!}{\rho^m} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) e^{-im\theta} \frac{d\theta}{2\pi}.$$

The obvious estimate

$$|f^{(m)}(z_0)| \leq \frac{m!}{\rho^m} \int_0^{2\pi} |f(z_0 + \rho e^{i\theta})| \frac{d\theta}{2\pi}$$

now leads to the following version of the Cauchy estimates.

Theorem (Cauchy Estimates). Suppose $f(z)$ is analytic for $|z - z_0| \leq \rho$. If $|f(z)| \leq M$ for $|z - z_0| = \rho$, then

$$\left| f^{(m)}(z_0) \right| \leq \frac{m!}{\rho^m} M, \quad m \geq 0.$$

Note that this estimate scales correctly with respect to M , in the sense that if we multiply $f(z)$ by a positive constant, the estimate is multiplied by the same constant. It also scales correctly with respect to ρ , in the sense that if we dilate the disk by a factor $c > 0$, then both the m th derivative of the dilated function and the factor $1/\rho^m$ are multiplied by $1/c^m$. The estimate is invariant under translations. In effect it would have sufficed to check the estimate in the special case of analytic functions on the unit disk that are bounded in modulus by one on the unit circle. In this special case the estimate asserts that all derivatives of the function are bounded in modulus by one at the origin.

As an application of the Cauchy estimates, we prove the following.

Theorem (Liouville's Theorem). Let $f(z)$ be an analytic function on the complex plane. If $f(z)$ is bounded, then $f(z)$ is constant.

Indeed, suppose $|f(z)| \leq M$ for all $z \in \mathbb{C}$. We can apply the Cauchy estimate to a disk centered at any z_0 , of any radius ρ , to obtain

$$|f'(z_0)| \leq \frac{M}{\rho}.$$

Letting ρ tend to $+\infty$, we obtain $f'(z_0) = 0$. Since this is true for all z_0 , $f(z)$ is constant.

We define an **entire function** to be a function that is analytic on the entire complex plane. The polynomials $a_n z^n + \cdots + a_1 z + a_0$ are entire functions. The transcendental functions e^z , $\cos z$, $\sin z$, $\cosh z$, $\sinh z$ are also entire. Any linear combination of entire functions is entire, and any product of entire functions is entire. Examples of functions that are not entire are $1/z$, $\log z$, and $\sqrt{z}$.

In terms of entire functions, Liouville's theorem has a succinct statement: *A bounded entire function is constant.*

As a test of the strength of Liouville's theorem, we apply it to give yet another proof (one of hundreds of proofs) of the fundamental theorem of algebra, that every polynomial in z of degree $n \geq 1$ has a zero. (See Section I.1.) The proof is by contradiction. Suppose $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ is a polynomial with no complex root. Then $1/p(z)$ is an entire function. Since

$$\frac{p(z)}{z^n} = 1 + \frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}$$

tends to 1 as $z \rightarrow \infty$, $p(z) \rightarrow \infty$ and $1/p(z) \rightarrow 0$ as $z \rightarrow \infty$. Consequently, $1/p(z)$ is bounded. By Liouville's theorem, $1/p(z)$ is constant. Since the constant cannot be 0, we have a contradiction. Our supposition is false, and $p(z)$ must have a zero.

Exercises for IV.5

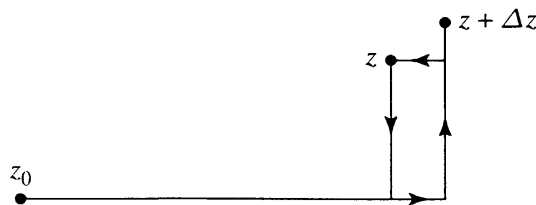
1. Show that if u is a harmonic function on $\mathbb{C}$ that is bounded above, then u is constant. *Hint.* Express u as the real part of an analytic function, and exponentiate.
2. Show that if $f(z)$ is an entire function, and there is a nonempty disk such that $f(z)$ does not attain any values in the disk, then $f(z)$ is constant.
3. A function $f(z)$ on the complex plane is **doubly periodic** if there are two periods ω_0 and ω_1 of $f(z)$ that do not lie on the same line through the origin (that is, ω_0 and ω_1 are linearly independent over the reals, and $f(z + \omega_0) = f(z + \omega_1) = f(z)$ for all complex numbers z). Prove that the only entire functions that are doubly periodic are the constants.
4. Suppose that $f(z)$ is an entire function such that $f(z)/z^n$ is bounded for $|z| \geq R$. Show that $f(z)$ is a polynomial of degree at most n . What can be said if $f(z)/z^n$ is bounded on the entire complex plane?
5. Show that if $\mathbf{V}(z)$ is the velocity vector field for a fluid flow in the entire complex plane, and if the speed $|\mathbf{V}(z)|$ is bounded, then $\mathbf{V}(z)$ is a constant flow.

6. Morera's Theorem

What Morera did was to observe that the differential $f(z)dz$ is closed if and only if $f(z)$ is analytic. The following more precise variant of this observation is often referred to as "Morera's theorem." It has a number of useful applications.

Theorem (Morera's Theorem). *Let $f(z)$ be a continuous function on a domain D . If $\int_{\partial R} f(z)dz = 0$ for every closed rectangle R contained in D with sides parallel to the coordinate axes, then $f(z)$ is analytic on D .*

The power of Morera's theorem resides in the fact that no hypothesis is made concerning the smoothness of $f(z)$. Only continuity is required.



To prove the theorem, we can assume that D is a disk with center z_0 . Define

$$F(z) = \int_{z_0}^z f(\zeta) d\zeta, \quad z \in D,$$

where the path of integration runs along a horizontal line and then a vertical line, as indicated in the figure. We could as well define $F(z)$ using the path starting from z_0 along a vertical line followed by a horizontal line. The hypothesis guarantees that these two paths yield the same integral, as the difference is the integral of $f(z)dz$ over the boundary of a rectangle. Now we differentiate $F(z)$ by hand. We have

$$F(z + \Delta z) - F(z) = \int_z^{z+\Delta z} f(\zeta) d\zeta,$$

where the path of integration is the path from z to $z + \Delta z$ that follows a horizontal line and then a vertical line, as in the figure. Here we have again used the fact that the integral along the boundary of a rectangle is zero, as indicated in the figure. To deal with the integral on the right, we use the trick of adding and subtracting $f(z)$ from the integrand. Since z is fixed, the value $f(z)$ is constant for the integration, and we obtain

$$\begin{aligned} F(z + \Delta z) - F(z) &= f(z) \int_z^{z+\Delta z} d\zeta + \int_z^{z+\Delta z} (f(\zeta) - f(z)) d\zeta \\ &= f(z)\Delta z + \int_z^{z+\Delta z} (f(\zeta) - f(z)) d\zeta. \end{aligned}$$

Now, the length of the contour from z to $z + \Delta z$ is at most $2|\Delta z|$. If we divide by Δz and use the ML -estimate on the last integral, we obtain

$$\left| \frac{F(z + \Delta z) - F(z)}{\Delta z} - f(z) \right| \leq 2M_\varepsilon, \quad |\Delta z| < \varepsilon,$$

where M_ε is the maximum of $|f(\zeta) - f(z)|$ over all ζ satisfying $|\zeta - z| \leq \varepsilon$. Since $f(\zeta)$ is continuous at z , M_ε tends to 0 as $\varepsilon \rightarrow 0$. Consequently, $F(z)$ is complex differentiable, with complex derivative $F'(z) = f(z)$. Since $f(z)$ is continuous, $F(z)$ is analytic, and since $f(z)$ is the derivative of an analytic function, $f(z)$ is also analytic.

There is a metatheorem to the effect that *if the integrand depends analytically on a parameter, then the integral depends analytically on the parameter*. This sort of theorem is easy to prove using Morera's theorem and